

# OC Core 1.3.1 Journal Core EN

## Document boundary

- This journal-core document is the reviewer and formal-reader route extracted from the real corpus.
- It prioritizes theorem scope, source audit, foundational consistency, and worked examples.
- The document is bounded by the same publication boundary as the master monograph.

## Contents

|          |                                                                             |           |
|----------|-----------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Reader Contract and Navigation Guide</b>                                 | <b>4</b>  |
| 1.1      | The five reading routes . . . . .                                           | 4         |
| 1.2      | Route legend . . . . .                                                      | 4         |
| 1.3      | External interaction crosswalk . . . . .                                    | 5         |
| 1.4      | What this guide is not . . . . .                                            | 5         |
| <b>2</b> | <b>Results and Derived Consequences</b>                                     | <b>6</b>  |
| 2.1      | Theorem 1: Monotonicity of dimensionality . . . . .                         | 6         |
| 2.2      | Theorem 2: Impossibility of spontaneous dimension creation . . . . .        | 6         |
| 2.3      | Theorem 3: Death through boundary and embedding collapse . . . . .          | 6         |
| 2.4      | Theorem 4: Impossibility of evolution outside the embedding space . . . . . | 7         |
| 2.5      | Theorem 5: Necessity and sufficiency of compatibility with $M$ . . . . .    | 7         |
| 2.6      | Theorem 6: Structural tension and phase transitions . . . . .               | 7         |
| 2.7      | Theorem 7: Universal law of complexity growth . . . . .                     | 8         |
| 2.8      | Theorem 8: Impossibility of eternal stabilisation . . . . .                 | 8         |
| 2.9      | Theorem 9: Death as loss of cycles . . . . .                                | 8         |
| 2.10     | Theorem 10: Monotonic growth of embedding spaces . . . . .                  | 9         |
| 2.11     | Theorem 11: Threshold–expressivity incompatibility . . . . .                | 9         |
| 2.12     | Theorem 12: Incompleteness of embedding spaces . . . . .                    | 9         |
| 2.13     | Corollaries and domain–independent consequences . . . . .                   | 9         |
| 2.14     | Implications for the vertical hierarchy . . . . .                           | 10        |
| 2.15     | Summary . . . . .                                                           | 10        |
| <b>3</b> | <b>Extended Boundary and Patch Geometry</b>                                 | <b>10</b> |
| 3.1      | Formal Definition of $\partial\Omega(K)$ . . . . .                          | 10        |
| 3.2      | Boundary Geometry . . . . .                                                 | 11        |
| 3.3      | Patch Model of Boundaries . . . . .                                         | 11        |
| 3.4      | Boundary Threshold System . . . . .                                         | 12        |
| 3.5      | Boundary–Driven Dynamics . . . . .                                          | 12        |
| 3.6      | Boundary Failure and Collapse . . . . .                                     | 13        |
| 3.7      | Boundary and Rebirth . . . . .                                              | 13        |

|          |                                                                                                |           |
|----------|------------------------------------------------------------------------------------------------|-----------|
| <b>4</b> | <b>Extended Threshold Landscape</b>                                                            | <b>14</b> |
| 4.1      | Definition of a Threshold . . . . .                                                            | 14        |
| 4.2      | Taxonomy of Thresholds . . . . .                                                               | 14        |
| 4.3      | Threshold Geometry . . . . .                                                                   | 15        |
| 4.4      | Dimensional Thresholds . . . . .                                                               | 16        |
| 4.5      | Threshold Cascades . . . . .                                                                   | 16        |
| 4.6      | Death Thresholds . . . . .                                                                     | 17        |
| 4.7      | Examples Across Levels . . . . .                                                               | 17        |
| 4.8      | Summary . . . . .                                                                              | 18        |
| <b>5</b> | <b>Module Architecture and Cross-Level Structures</b>                                          | <b>18</b> |
| <b>6</b> | <b>Source Witnessing, Theorem Fate, and Publication Boundary</b>                               | <b>19</b> |
| 6.1      | Canonical scientific authority . . . . .                                                       | 19        |
| 6.2      | Source witnessing . . . . .                                                                    | 19        |
| 6.3      | Theorem fate and scope discipline . . . . .                                                    | 20        |
| 6.4      | Relation to journal-core and companion artifacts . . . . .                                     | 20        |
| 6.5      | Downstream projection rule . . . . .                                                           | 21        |
| <b>7</b> | <b>Core 1.3 Foundational Consistency Dossier</b>                                               | <b>22</b> |
| 7.1      | Why the hierarchy extends through $K_{12}$ . . . . .                                           | 22        |
| 7.2      | Scientific role of the upper levels . . . . .                                                  | 22        |
| 7.3      | Statement classes . . . . .                                                                    | 22        |
| 7.4      | Proof-bearing and nonclaim layers . . . . .                                                    | 24        |
| 7.5      | Relation between master monograph and journal core . . . . .                                   | 24        |
| 7.6      | Reader-facing consequence . . . . .                                                            | 25        |
| <b>8</b> | <b>Theorem Roadmap and Proof Navigation</b>                                                    | <b>26</b> |
| 8.1      | Load-bearing theorem spine . . . . .                                                           | 26        |
| 8.2      | What surrounds the spine . . . . .                                                             | 27        |
| 8.3      | Statement classes and burden . . . . .                                                         | 28        |
| 8.4      | Shortest hostile-review checklist . . . . .                                                    | 28        |
| <b>9</b> | <b>Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks</b>               | <b>30</b> |
| 9.1      | Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model . . . . . | 30        |
| 9.2      | Worked Example 2: why $K_{11}$ and $K_{12}$ are not decorative . . . . .                       | 32        |
| 9.3      | Interpretive Check 3: how the journal core is lawfully extracted from the monograph . . . . .  | 35        |
| 9.4      | What would count as a real counterexample . . . . .                                            | 35        |
| 9.5      | Why examples belong in a strict scientific monograph . . . . .                                 | 35        |
| <b>A</b> | <b>Core 1.3 Source-Audit Appendix</b>                                                          | <b>37</b> |
| A.1      | Why an appendix is still needed . . . . .                                                      | 37        |
| A.2      | Witness classes . . . . .                                                                      | 37        |
| A.3      | Questions that the appendix answers . . . . .                                                  | 37        |
| A.4      | Current boundary . . . . .                                                                     | 37        |
| <b>B</b> | <b>Journal-Core Extraction Bridge</b>                                                          | <b>37</b> |
| B.1      | Why the journal-core exists . . . . .                                                          | 38        |
| B.2      | What the journal-core may omit . . . . .                                                       | 38        |
| B.3      | What the monograph guarantees to later artifacts . . . . .                                     | 38        |

|          |                                                                |           |
|----------|----------------------------------------------------------------|-----------|
| <b>C</b> | <b>Reviewer Navigation Matrix and Audit Checklist</b>          | <b>39</b> |
| C.1      | Review questions, compact route, and decisive checks . . . . . | 39        |
| C.2      | Compact reading order by reviewer role . . . . .               | 40        |
| <b>D</b> | <b>OC 1.3.1 Science Delta Appendix</b>                         | <b>41</b> |
| <b>E</b> | <b>OC 1.3.1 Critique Closure Appendix</b>                      | <b>41</b> |

# 1 Reader Contract and Navigation Guide

Core 1.3 is the flagship reader-facing monograph, but readers arrive with different tasks. This guide is therefore a route map rather than a second introduction. Its job is to show where each of the five reader roles should begin: Reviewer route, Formal reader route, Domain reader route, Broad scientific reader route, and Practical user route. External collaborators who need deployment guidance should use the Practical user route.

The dedicated Figure Atlas in Appendix ?? and the Technical Derivation Atlas in Appendix ?? are intentionally outside the default continuous reading route. Their material belongs to the appendix inspection layer; the formal-reader route below consults the Technical Derivation Atlas only when local proof support is needed.

## 1.1 The five reading routes

The reviewer-navigation appendix uses slightly narrower review labels: “rapid critical reviewer” corresponds to the Reviewer route, “formal mathematical reviewer” to the Formal reader route, and “generalist scientific reviewer” to the Broad scientific reader route.

**Reviewer route.** Read the front matter, this guide, Sections 2, 6–8, Section 9, and then Appendices A–C. That route exposes the publication boundary, theorem status, load-bearing theorem spine, reviewer objections, and source witnesses as quickly as possible.

**Formal reader route.** Use Sections ??–?? for orientation only. Then read Sections 3–??, Section 5, and Sections 6–?? in order. Use Appendices ??–??, Appendix ??, and the Technical Derivation Atlas in Appendix ?? when local proof support is needed. This is the compact route that keeps exact doctrine, theorem scope, and proof machinery visible at once.

**Domain reader route.** Read Section ?? and Sections 9–??, then move into Appendices ??, ??, ??, ??, ??, and ??, with Appendix ?? added only when experiments, predictions, or falsifiability catalogues need inspection. That path exposes bounded domain embedding, worked interpretation, empirical promotion rules, replay surfaces, benchmark cases, execution board, practical use, and comparison evidence without forcing the reader to traverse every technical-derivation section first.

**Broad scientific reader route.** Read Sections ??–??, Sections 3–??, Section 5, and Sections 6–?? straight through as a continuous graduate-level monograph. Use Appendix ?? for notation when needed and defer Appendices ??–?? until a concrete figure, benchmark table, or support dossier question arises. This is the recommended route for a broad scientific reader.

**Practical user route.** Read the front matter, this guide, Section 6, Section ??, and Appendices A, B, and ?? first; add Appendix ?? only when the task requires experiments, predictions, or falsifiability catalogues. Return to Sections ??–?? only when a practical use case needs theorem, replay, or falsifier support to be inspected in detail. This route is for external collaborators who need deployment guidance: what can be predicted, which decisions can be supported, which procedures can be run, and how the result compares with alternatives.

## 1.2 Route legend

The five routes above are the working navigation system:

- **Reviewer route:** shortest path to claim boundary, proof spine, objections, and source witnesses.
- **Formal reader route:** compact path through doctrine, theorem scope, notation, and proof machinery.
- **Domain reader route:** bounded path from domain embedding to replay, benchmarks, execution, use, and comparison evidence.
- **Broad scientific reader route:** continuous graduate-level reading path through the main argument before atlas inspection.
- **Practical user route:** shortest path to usable tasks, required inputs, outputs, comparison boundaries, and deployment constraints.

### 1.3 External interaction crosswalk

The crosswalk maps common external needs to the shortest route bundle that can answer them while keeping the supporting audit trail visible.

| External task                     | Primary route                               | Support route if verification is needed                                                                                                                                         |
|-----------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evaluate the release for review   | This guide; Sections 2, 6–9; Appendices A–C | Appendix ?? when proof-obligation details are needed                                                                                                                            |
| Audit the bounded theorem core    | Section 2, Section 7, and Section 8         | Appendix ??, Appendix ??, Appendix C, and Appendix ??                                                                                                                           |
| Check the empirical promotion bar | Sections ??–??                              | Appendix ??, Appendix ??, Appendix ??, Appendix ??, Appendix ??, and Appendix ??                                                                                                |
| Cite the unified synthesis        | Section ??                                  | Appendix ??, Appendix ??, and the K-level parameter tables in Appendix ??                                                                                                       |
| Use or compare the model          | Section ??                                  | First inspect Appendix ??; use Appendix ?? only for experiments, predictions, or falsifiability catalogues; then use Sections ??–?? only when proof or replay support is needed |
| Verify publication boundaries     | Section 6                                   | Appendix A and Appendix B                                                                                                                                                       |

Table 1: Stable reading routes for the most common external interactions. The purpose is to keep external use concise while preserving the audit trail behind each cited claim.

### 1.4 What this guide is not

This guide does not restate the theory, replace the theorem roadmap, or turn the flagship monograph into a status summary. It simply tells the reader where each task belongs. Section 8 covers proof navigation. Section ?? covers the unified synthesis. Section ?? covers practical consequences and comparative analysis. The appendices keep the inspection burden inside the same volume without displacing the main reading spine.

## 2 Results and Derived Consequences

This section summarises the core structural results that follow from the formal framework introduced in Section ???. It presents the canonical consequences of the Ontology of Continua (OC) axioms and operators in their Core 1.3 form.

All results in this chapter are domain-independent. They follow solely from the structural definitions of continua, axes, thresholds, potentials, flows, boundaries, cycles and the measure of continuumness  $k(K, t)$ . Proofs are not reproduced here in full detail; they are deferred to dedicated extension papers. The focus is on stating the key theorems and consequences that constrain all continua in the hierarchy  $K_0$ – $K_{12}$ .

### 2.1 Theorem 1: Monotonicity of dimensionality

**Statement.** If a continuum  $K_{x+1}$  emerges from  $K_x$ , then

$$\dim(K_{x+1}) > \dim(K_x).$$

**Justification (sketch).** Dimensional emergence is defined as the appearance of a new axis  $A_{\text{new}}$  that is incompatible with all existing axes in  $A(K_x)$ ; that is, it cannot be represented as a linear or structural combination of them. The dimensional threshold  $\Theta_{\dim}(K_x)$  enforces that  $A_{\text{new}}$  is nondegenerate and structurally necessary. Hence, any emergent continuum has strictly higher dimensionality than its predecessor.

**Corollary 1.1 (No degradational simplification).** There is no continuum-level evolution that reduces dimensionality while preserving existence: if  $K(t)$  is live ( $k(K, t) > 0$ ), then

$$\dim(K(t + dt)) \geq \dim(K(t)).$$

Apparent reduction of dimension corresponds to the death of the higher-dimensional continuum and the birth of a different, lower-dimensional one, not to a continuous simplification.

### 2.2 Theorem 2: Impossibility of spontaneous dimension creation

**Statement.** No continuum can increase its dimensionality without exceeding the dimensional threshold and having access to a suitable axis in its embedding space:

$$\dim(K_{x+1}) > \dim(K_x) \quad \Rightarrow \quad T(K_x, t) > \Theta_{\dim}(K_x) \wedge A_{\text{new}} \in A(M_x) \setminus A(K_x).$$

**Justification (sketch).** New axes represent classes of differences that are incompatible with existing axes. Such incompatibility arises only when structural tension  $T(K_x, t)$  with respect to the current threshold landscape cannot be resolved within the existing axes. By definition of  $\Theta_{\dim}$ , below this threshold all differences can be projected onto the structural span  $\text{span}_{\text{str}}(A(K_x))$ ; above it, projection fails and a new axis is required. The new axis must already be available in the embedding space  $M_x$ ; otherwise there is no structural direction along which the continuum can expand.

**Consequence 2.1.** Noise, local fluctuations or internal flows that do not raise structural tension above  $\Theta_{\dim}(K_x)$  cannot generate new dimensions.

### 2.3 Theorem 3: Death through boundary and embedding collapse

**Statement.** A continuum  $K$  dies when its admissible state space collapses:

$$\Omega(K) = \emptyset.$$

**Equivalent formulations.** The following conditions are equivalent and characterise the death of  $K$ :

1. supporting cycles disappear,  $C(K) = \emptyset$ ;

2. supporting flows vanish,  $J_{\text{support}} = 0$ , or are insufficient to maintain any cycles;
3. there exists a state-independent violation of thresholds,  $\forall s : \exists k \text{ with } f_k(s) > 0$ ;
4. continuumness collapses,  $k(K, t) \rightarrow 0$ ;
5. the continuum exits the region supported by its embedding space  $M$ , i.e. no state satisfies both the internal thresholds of  $K$  and the external constraints of  $M$ .

**Corollary 3.1 (Death as exit from embedding space).** If the constraints of  $M$  change so that no configuration of  $K$  can be embedded into  $M$  while satisfying its thresholds, then  $\Omega(K)$  becomes empty and the continuum dies, regardless of its internal structure.

**Corollary 3.2 (Irreversibility of death).** Once  $\Omega(K) = \emptyset$ , no operator  $E$  acting within the same level, nor any interaction operator  $E_{\text{int}}$  that preserves the identity of  $K$ , can reconstruct a nonempty  $\Omega(K)$ . Any apparent “recovery” corresponds to the birth of a new continuum  $K'$ , not the resurrection of the original  $K$ .

## 2.4 Theorem 4: Impossibility of evolution outside the embedding space

**Statement.** Let  $K_x$  be a continuum embedded in  $M_x$ . Then for all times  $t$ ,

$$A(K_x(t)) \subseteq A(M_x), \quad \text{span}_{\text{str}}(A(K_x(t + dt))) \subseteq \text{span}_{\text{str}}(A(M_x)).$$

**Consequence 4.1.** All dynamical processes of  $K_x$ , including dimensional birth  $K_x \rightarrow K_{x+1}$ , require that the embedding space  $M_x$  already contain the axes along which the evolution proceeds. A continuum cannot evolve in directions not present in its embedding space.

## 2.5 Theorem 5: Necessity and sufficiency of compatibility with $M$

**Statement.** A continuum  $K$  exists as a live continuum (i.e.  $\Omega(K) \neq \emptyset$  and  $k(K, t) > 0$ ) if and only if it is compatible with its embedding space  $M$ :

$$K \text{ exists} \iff \Omega(K) \neq \emptyset \wedge A(K) \subseteq A(M) \wedge \Theta(K) \text{ is satisfiable within } M.$$

**Consequence 5.1.** Even if a configuration is mathematically well-defined at the level of  $K$ , it does not correspond to a live continuum unless the embedding space can support the required axes, thresholds and flows. Incompatible configurations are excluded structurally by setting  $\Omega(K) = \emptyset$ .

## 2.6 Theorem 6: Structural tension and phase transitions

**Statement.** Let  $T(K, t)$  denote structural tension, understood as a functional of potentials, axes and their gradients:

$$T(K, t) = T(P(t), A, \nabla P(t)).$$

Then:

- a phase transition occurs when  $T(K, t) = \Theta_{\text{crit}}(K)$ ,
- a dimensional transition occurs when  $T(K, t) = \Theta_{\text{dim}}(K)$ ,
- structural collapse occurs when  $T(K, t) = \Theta_{\text{death}}(K)$ .

**Corollary 6.1.** All qualitative changes in the continuum — emergence of new regimes, new dimensions, or collapse — are governed by the relationship between structural tension and the threshold landscape.

## 2.7 Theorem 7: Universal law of complexity growth

**Statement.** For any live continuum  $K$ ,

$$S(K(t + dt)) \geq S(K(t)),$$

where  $S$  is a structural measure of complexity defined on the state space and its organisation (for example, combining contributions from number of axes, diversity of states, richness of cycles and properties of the embedding).

**Consequence 7.1.** Complexity increases strictly whenever:

- new axes appear;
- new stable cycles form;
- the admissible state space  $\Omega(K)$  expands;
- the embedding space  $M$  gains new axes relevant to  $K$ .

Stagnant or constant complexity corresponds to finely balanced dynamics below critical thresholds; however, this regime is generically unstable (see Theorem 2.8).

## 2.8 Theorem 8: Impossibility of eternal stabilisation

**Statement.** There is no nontrivial live continuum  $K$  that remains forever at a fixed point in its state space while maintaining positive continuumness. Formally, there is no solution with

$$\frac{dP}{dt} = 0, \quad \frac{dJ}{dt} = 0, \quad \frac{d\Theta}{dt} = 0, \quad \frac{dk}{dt} = 0$$

for all  $t$ , except the trivial case where  $K$  is structurally frozen and decoupled from its embedding space.

**Justification (sketch).** Embedding spaces  $M_x$  themselves evolve and expand (Theorem 10). Internal and external flows cannot both be identically zero on all time scales without either:

- driving the continuum to collapse (loss of supporting flows), or
- inducing structural changes (axes, thresholds, cycles).

Hence any nontrivial live continuum is subject to evolution of its structure or to eventual death.

## 2.9 Theorem 9: Death as loss of cycles

**Statement.** A live continuum  $K$  dies when its maximal structurally stable cycle complex disappears:

$$C_{\max}(K) = \emptyset.$$

**Justification (sketch).** Cycles represent self-maintaining flows that keep the continuum away from its boundary  $\partial\Omega$ . If no such cycle exists, then:

- supporting flows cannot form closed loops;
- trajectories either approach  $\partial\Omega$  or violate thresholds;
- continuumness  $k(K, t)$  decays to zero.

Therefore loss of the maximal cycle complex coincides with the collapse of  $\Omega(K)$ .

**Corollary 9.1.** Death can be detected structurally by the disappearance of all cycles satisfying a minimal stability condition (strictly positive distance to  $\partial\Omega$ ).



## 2.10 Theorem 10: Monotonic growth of embedding spaces

**Statement.** Embedding spaces form a monotonic sequence:

$$M_0 \subset M_1 \subset \dots \subset M_x \subset \dots,$$

and whenever a new continuum  $K_{x+1}$  emerges, the corresponding embedding space  $M_{x+1}$  strictly extends  $M_x$  in terms of its available axes:

$$A(M_x) \subset A(M_{x+1}).$$

**Consequence 10.1.** The growth of continua in dimension and complexity is inseparable from the growth of their embedding spaces. New continua cannot appear without an expansion of the structural possibilities encoded in  $M$ .

## 2.11 Theorem 11: Threshold–expressivity incompatibility

**Statement.** A continuum  $K$  collapses when its axes become insufficient to express the required differences:

$$\dim(A(K)) < \dim(\text{Differences}(K)),$$

where  $\text{Differences}(K)$  is the effective dimension of structurally relevant distinctions imposed by the embedding space and thresholds.

**Consequence 11.1.** Collapse may occur even without violating energetic or dynamical constraints if representational capacity is exceeded. In particular, cognitive, social or meta–theoretical continua can die purely due to loss of expressive adequacy of their axes.

## 2.12 Theorem 12: Incompleteness of embedding spaces

**Statement.** For any finite embedding space  $M_x$ , there exist potential continua  $K'$  that cannot be realised within  $M_x$  because their axes or thresholds are incompatible with  $A(M_x)$  or with the constraints of  $M_x$ .

**Consequence 12.1.** No single embedding space  $M_x$  is universal for all possible continua. The sequence of embedding spaces must itself expand and diversify to host new structural regimes.

## 2.13 Corollaries and domain–independent consequences

Collecting the theorems above, we obtain the following domain–independent consequences:

- Dimensionality is strictly monotonic for live continua; no continuous degradation of dimension is possible.
- Dimension cannot appear spontaneously; it requires structural tension above  $\Theta_{\text{dim}}$  and the presence of suitable axes in the embedding space.
- Death is characterised by the collapse of  $\Omega(K)$ , the disappearance of cycles and the irreversibility of this collapse.
- Evolution is constrained by embedding spaces; continua cannot evolve in directions that  $M$  does not support.
- Complexity tends to grow for live continua, driven by new axes, new cycles and expanding state spaces.
- Eternal exact stabilisation is structurally excluded for nontrivial continua; they either evolve or die.
- Representational and expressive limits can cause collapse even when energetic and dynamical limits are not yet reached.

These consequences hold uniformly for all levels  $K_0$ – $K_{10}$ ; their domain–specific content arises only from the interpretation of axes, potentials, thresholds and flows in each level.

## 2.14 Implications for the vertical hierarchy

In the vertical hierarchy  $K_0$ – $K_{10}$ , the structural theorems manifest as follows:

- $K_0$ : no dynamics, no birth or death; only structural conditions for distinguishability.
- $K_1$ : classical phase-like behaviour with continuous flows and basic stability/critical thresholds.
- $K_2$ : physical thresholds, including percolation, BKT transitions, coherence/condensation thresholds and mass-generating mechanisms.
- $K_3$ – $K_4$ : chemical and prebiotic thresholds (RAF closure, membrane formation, osmotic and curvature thresholds, redox and pH thresholds).
- $K_5$ : electrical thresholds for excitation, spiking and gradient-driven flows in early neural systems.
- $K_6$ : logical and representational thresholds for cognitive binding, prediction, memory stability and internal model coherence.
- $K_7$ – $K_8$ : social and civilizational thresholds governing trust, institutional stability, systemic resilience and collapse.
- $K_9$ – $K_{10}$ : expressive, coherence and consistency thresholds in the space of theories, paradigms, ontologies and meta-theoretical structures.

In each case, the same structural principles — monotonic dimension, threshold-governed emergence and collapse, dependence on embedding spaces, and complexity growth — apply, while the concrete interpretation of variables changes.

## 2.15 Summary

The results presented in this chapter constitute the structural backbone of OC. They state that dimension is monotonic, that emergent dimensions require threshold-induced phase transitions, that death is characterised by the collapse of admissible states and is irreversible, that evolution is constrained by embedding spaces, that complexity tends to grow for live continua, that cycles are the carriers of structural life, and that expressive limits can cause collapse. These principles apply across the entire continuum hierarchy and define the constraints under which all continua can exist, evolve or die.

# 3 Extended Boundary and Patch Geometry

This chapter restores the full theory of boundaries and patch geometry. Boundaries constitute one of the most universal structures in the OC framework: they govern phase separation in  $K_2$ , membrane formation in  $K_3$ – $K_4$ , excitability in  $K_5$ , and representational or institutional limits in  $K_6$ – $K_8$ . The present version consolidates the complete structure needed for Core 1.3: threshold-defined boundaries, patch discretisation, local threshold classes, and boundary-driven transitions.

## 3.1 Formal Definition of $\partial\Omega(K)$

For any continuum  $K$  with admissible state space  $\Omega(K)$  and threshold landscape  $\Theta(K)$ , the boundary is defined as the set of states at which at least one threshold is exactly saturated:

$$\partial\Omega(K) = \{ s \in \overline{\Omega(K)} : \exists k \text{ such that } g_k(s) = \Theta_k \}.$$

Equivalently, if  $f_k(s) = g_k(s) - \Theta_k$  denotes the shifted residual form used in the notation appendix and the model chapter, then interior states satisfy  $f_k(s) < 0$ , admissible-side states satisfy  $f_k(s) \leq 0$ , and the same boundary is the zero set  $f_k(s) = 0$  for at least one active threshold.

Thus:

- interior states satisfy  $f_k(s) < \Theta_k$  for all thresholds;
- exterior states violate at least one threshold;
- the boundary is a *threshold-defined surface*, not a geometric primitive.

This definition unifies classical surfaces, membranes, vesicles, percolation fronts, excitable boundaries, logical limits and institutional thresholds under one structure.

### 3.2 Boundary Geometry

Boundary geometry emerges from the local structure of thresholds, potentials and flows. If the active threshold at a point  $s \in \partial\Omega(K)$  is  $g_k$ , the outward normal is aligned with the gradient:

$$n(s) \propto \nabla g_k(s).$$

Multiple active thresholds create corners, ridges and high-curvature regions. Curvature becomes dynamically relevant whenever bending or surface-energy terms enter the potential:

$$P_{\text{bend}}(s) = \frac{1}{2} \kappa(s)^2 \kappa_b,$$

with bending rigidity  $\kappa_b$ .

This connects:

- $K_2$ : classical surface tension,
- $K_3$ – $K_4$ : membrane curvature and vesicle energetics,
- $K_5$ : excitable boundary geometry.

### 3.3 Patch Model of Boundaries

OC models boundaries not as continuous uniform surfaces but as collections of dynamic *patches*:

$$\partial\Omega(K) = \bigcup_i \partial\Omega_i.$$

Each patch  $P_i$  carries:

$$(\sigma_i, \Theta_i, P_i(t), J_i(t)),$$

representing its local state, thresholds, potentials and flows.

**Local states.** Patches take symbolic states:

$$\sigma_i \in \{L_\alpha, L_\beta, L_o, L_f, L_b\},$$

encoding local packing or structural regimes:

- $L_\alpha$ : fluid-disordered,
- $L_\beta$ : ordered/gel,
- $L_o$ : liquid-ordered,
- $L_f$ : fragile/thin (pre-rupture),
- $L_b$ : broken boundary state.

This alphabet generalises beyond biological membranes to percolation edges, fracture fronts, fluctuating interfaces and even symbolic/cognitive boundaries.

**Local thresholds.** Each patch has a vector:

$$\Theta_i = (\Theta_i^{\text{perm}}, \Theta_i^{\text{grad}}, \Theta_i^{\text{mem}}, \Theta_i^{\text{bend}}, \dots),$$

with corresponding local potentials  $P_i(t)$  and flows  $J_i(t)$ .

### 3.4 Boundary Threshold System

Across  $K_2$ – $K_5$  three threshold classes appear universally.

**1. Permeability threshold  $\Theta_{\text{perm}}$ .** A patch becomes permeable when:

$$|\nabla P_i| > \Theta_i^{\text{perm}}.$$

Examples:

- osmotic permeability ( $K_3$ – $K_4$ ),
- ion leak onset ( $K_5$ ),
- phase–front penetration ( $K_2$ ).

**2. Gradient threshold  $\Theta_{\text{grad}}$ .** If a gradient is unsustainable:

$$|\nabla P_i| > \Theta_i^{\text{grad}} \Rightarrow \sigma_i \rightarrow L_f.$$

This governs:

- vesicle rupture at high osmotic pressure ( $K_4$ ),
- action–potential initiation at axon initial segments ( $K_5$ ),
- accelerated phase fronts and crack propagation ( $K_2$ ).

**3. Membrane integrity threshold  $\Theta_{\text{mem}}$ .** If membrane tension or curvature exceeds limits:

$$\gamma_i > \Theta_i^{\text{mem}} \quad \text{or} \quad \kappa(s) > \Theta_i^{\text{bend}},$$

then:

$$\sigma_i \rightarrow L_b.$$

This yields a unified understanding of rupture, buckling, pore formation, boundary breakdown and surface collapse.

### 3.5 Boundary–Driven Dynamics

Neighbouring patches interact via flows:

$$J_{ij} = F(P_i, P_j, \sigma_i, \sigma_j).$$

This produces:

- stabilisation and rearrangement of patch mosaics,
- propagation of rupture fronts,
- curvature–driven migration,
- pore initiation,
- excitable–medium behaviour in  $K_5$ .

Patch evolution follows:

$$\frac{d\sigma_i}{dt} = S(\sigma_i, P_i, J_i, \Theta_i),$$

where  $S$  is a structural update operator independent of domain.

**Boundary Localisation Principle.** *All  $K_3$ – $K_5$  dimensional transitions initiate on boundary patches.*

This explains:

- protocell bursting beginning at high-curvature points,
- action potentials initiating at axon initial segments,
- phase transitions nucleating at defects or surfaces.

### 3.6 Boundary Failure and Collapse

Boundary collapse occurs when a connected set of patches becomes broken:

$$\sigma_i = L_b \quad \forall i \in \mathcal{C},$$

and the cluster  $\mathcal{C}$  percolates across the boundary.

By Theorem 3 (Results chapter),

$$\partial\Omega(K) \text{ collapses} \implies \Omega(K) = \emptyset,$$

and the continuum dies.

We distinguish:

- **local failure:** isolated  $L_b$  patches, reversible;
- **cluster collapse:** percolating  $L_b$  network, irreversible;
- **global boundary loss:** disappearance of the boundary, implying death or transition to a new continuum.

### 3.7 Boundary and Rebirth

Birth of a new continuum  $K_{x+1}$  typically requires:

- formation or rupture of boundary patches,
- creation of new patch types,
- emergence of new threshold classes on boundary regions,
- establishment of a new global boundary geometry.

**Boundary Rebirth Principle.** *New continua emerge from new boundary geometries.*

Examples:

- membrane closure in the  $K_3 \rightarrow K_4$  transition,
- appearance of excitable boundaries in  $K_4 \rightarrow K_5$ ,
- formation of institutional boundaries in  $K_6 \rightarrow K_7$ ,
- emergence of representational boundaries in  $K_9 \rightarrow K_{10}$ .

Boundaries are therefore not merely limits: they are generative surfaces where new axes, constraints and cycles are born.

## 4 Extended Threshold Landscape

This chapter reconstructs the full theory of thresholds in the Ontology of Continua (OC), consolidating structural elements from the Physics, Chemistry, Biology and Cognition runs. Thresholds are the central mechanism through which qualitative change, dimensional emergence, collapse and death are produced.

Thresholds are universal: they apply to physical, chemical, biological, cognitive, social and theoretical continua throughout the hierarchy  $K_0$ – $K_{10}$ .

### 4.1 Definition of a Threshold

A *threshold* is represented by a constraint function

$$f_k : \Omega(K) \rightarrow \mathbb{R}$$

with the property

$$f_k(s) \leq 0 \quad \text{for all admissible states } s \in \Omega(K).$$

Thresholds separate qualitatively different dynamical regimes:

- $f_k(s) < 0$ : safe regime (interior),
- $f_k(s) = 0$ : boundary regime,
- $f_k(s) > 0$ : forbidden regime (outside  $\Omega(K)$ ).

Thus thresholds jointly determine:

- the shape of the admissible set  $\Omega(K)$ ,
- the boundary  $\partial\Omega(K)$ ,
- the conditions under which the continuum exists or collapses.

Thresholds are not static constants. They depend on potentials  $P(t)$ , flows  $J(t)$ , axes  $A(K)$ , environmental constraints and the structure of the embedding space  $M$ . At the structural level this is encoded by the threshold–evolution operator  $H$  from the general evolution machinery:

$$\frac{d\Theta_k}{dt} = H_k(\Theta_k, P, J, A, M),$$

where  $\Theta_k$  denotes the parameter set defining the constraint  $f_k$ .

### 4.2 Taxonomy of Thresholds

OC distinguishes seven universal classes of thresholds. This taxonomy was stabilised in Core 2.x and clarified across the domain runs.

**1. Existence thresholds**  $\Theta_{\text{exist}}$ . Conditions required for  $\Omega(K) \neq \emptyset$ . Examples:

- minimal energy conditions for bound physical states ( $K_2$ ),
- minimal catalytic closure (RAF existence) in  $K_3$ ,
- minimal membrane continuity and integrity in  $K_4$ ,
- minimal coherence of internal models in  $K_6$ ,
- minimal legitimacy and trust for institutions in  $K_7$ .

**2. Stability thresholds  $\Theta_{\text{stab}}$ .** Conditions under which flows remain bounded and cycles are stable. Examples:

- confinement and boundedness conditions in physics ( $K_2$ ),
- osmotic homeostasis conditions in protocells ( $K_4$ ),
- membrane potential stability in neurons ( $K_5$ ),
- normative stability in social continua ( $K_7$ ).

**3. Critical thresholds  $\Theta_{\text{crit}}$ .** Surfaces of qualitative change at fixed dimensionality. Examples:

- phase transitions and BKT-type thresholds ( $K_2$ ),
- onset of catalytic networks ( $K_3$ ),
- metabolic switching in  $K_4$ ,
- spike initiation threshold in  $K_5$ ,
- prediction–error bifurcations in  $K_6$ ,
- institutional bifurcations in  $K_7$ .

**4. Dimensional thresholds  $\Theta_{\text{dim}}$ .** Constraints that trigger the creation of new axes. Dimensional thresholds are central in OC:

$$T(K, t) > \Theta_{\text{dim}}(K) \Rightarrow \text{emergence of a new axis } A_{\text{new}}.$$

They govern the transitions  $K_x \rightarrow K_{x+1}$  throughout the hierarchy.

**5. Death thresholds  $\Theta_{\text{death}}$ .** Values beyond which no admissible state remains. Equivalently,

$$\forall s \in \overline{\Omega(K)} : \exists k \text{ with } f_k(s) > 0 \iff \Omega(K) = \emptyset.$$

**6. Expressivity thresholds  $\Theta_{\text{expr}}$ .** Thresholds on representational capacity. A continuum collapses when

$$\dim(A(K)) < \dim(\text{Differences}(K)),$$

where  $\text{Differences}(K)$  denotes the effective dimension of structurally relevant distinctions imposed by the embedding space and constraints. This class is crucial for cognitive, social and meta–theoretical collapse.

**7. Embedding thresholds  $\Theta_{\text{embed}}$ .** Constraints imposed by the embedding space  $M_x$ . A continuum cannot exist as a live continuum if the embedding space does not support its axes, thresholds or flows; formally this is captured by the compatibility conditions in Theorems 4 and 5 (Results chapter).

### 4.3 Threshold Geometry

Thresholds induce a stratified geometry on  $\Omega(K)$ . For each threshold function  $f_k$  we consider the level sets

$$\Sigma_k(c) = \{s \in \Omega(K) \mid f_k(s) = c\}.$$

In particular:

- $\Sigma_k(0)$  defines components of the boundary  $\partial\Omega(K)$ ,
- $\Sigma_k(c < 0)$  defines safe regions,
- $\Sigma_k(c > 0)$  defines forbidden regions.

Interactions between thresholds generate:

- multi-threshold intersections,
- cusp and fold structures (catastrophe-like geometries),
- regions of concentrated curvature,
- patch-level heterogeneity at the boundary (cf. Section 3).

The joint threshold geometry determines:

- the global shape of  $\Omega(K)$ ,
- admissible trajectories and cycle structure,
- the available transitions of the continuum.

#### 4.4 Dimensional Thresholds

Dimensional thresholds occupy a privileged role in OC because they decide when new differences become unrepresentable within the current axes.

Let  $T(K, t)$  denote structural tension. Then, schematically:

$$T(K, t) < \Theta_{\dim}(K) \Rightarrow \text{all relevant differences can be expressed within } A(K),$$

$$T(K, t) = \Theta_{\dim}(K) \Rightarrow \text{projection onto } A(K) \text{ fails,}$$

$$T(K, t) > \Theta_{\dim}(K) \Rightarrow \text{a new axis } A_{\text{new}} \text{ is structurally required.}$$

Dimensional thresholds govern, in particular:

- birth of  $K_1$  from the structural substrate  $K_0$ ,
- emergence of chemical axes in  $K_3$ ,
- membrane axes in  $K_4$ ,
- excitation axes in  $K_5$ ,
- cognitive representational axes in  $K_6$ ,
- social normative axes in  $K_7$ ,
- meta-logical axes in  $K_9$ – $K_{10}$ .

#### 4.5 Threshold Cascades

Thresholds frequently interact in cascades of the form

$$\Theta_{\text{grad}} \longrightarrow \Theta_{\text{perm}} \longrightarrow \Theta_{\text{mem}},$$

or more generally, gradients  $\rightarrow$  permeability  $\rightarrow$  structural failure.

Representative examples:

- **Vesicle bursting** ( $K_4$ ). Osmotic gradients exceed  $\Theta_{\text{grad}}$ , permeability increases ( $\Theta_{\text{perm}}$ ), and membrane integrity is lost ( $\Theta_{\text{mem}}$ ).
- **Action potentials** ( $K_5$ ). Depolarisation exceeds a gradient threshold, channels open at  $\Theta_{\text{crit}}$ , and a spike cycle is activated.
- **Institutional collapse** ( $K_7$ ). Trust deficits exceed  $\Theta_{\text{grad}}$ , normative coherence fails ( $\Theta_{\text{expr}}$ ), and institutional death follows ( $\Theta_{\text{death}}$ ).

Such cascades are central for early life, neural systems, and social and infrastructural dynamics.



## 4.6 Death Thresholds

Death thresholds determine when a continuum ceases to exist as a live continuum. The following structural conditions are equivalent manifestations of death:

- complete violation of thresholds at all states:  $\forall s : \exists k$  with  $f_k(s) > 0$ ;
- disappearance of supporting cycles:  $C(K) = \emptyset$ ;
- collapse of the admissible state space:  $\Omega(K) = \emptyset$ ;
- representational failure, e.g.  $\Theta_{\text{expr}}$  exceeded;
- incompatibility with the embedding space (no configuration satisfies both internal and embedding constraints).

Once  $\Omega(K) = \emptyset$ , death is structurally irreversible: no operator acting within the same level can recreate  $\Omega(K)$ ; any apparent recovery corresponds to the birth of a new continuum  $K'$ .

## 4.7 Examples Across Levels

The threshold taxonomy manifests differently at each level of the hierarchy.

### $K_1$ (Geometric continua).

- $\Theta_{\text{exist}}$ : differentiability and regularity conditions defining the classical region  $\Omega_{\text{cl}}$ ,
- $\Theta_{\text{crit}}$ : classical bifurcations and loss of stability.

### $K_2$ (Physical continua).

- $\Theta_{\text{crit}}$ : phase transitions, including BKT thresholds,
- $\Theta_{\text{dim}}$ : coherence thresholds associated with mass generation and ordering,
- $\Theta_{\text{death}}$ : confinement/decoupling limits where the relevant continuum ceases to exist.

### $K_3$ (Chemical continua).

- $\Theta_{\text{exist}}$ : catalytic closure and RAF existence,
- $\Theta_{\text{crit}}$ : reaction–network percolation,
- $\Theta_{\text{expr}}$ : insufficient chemical axes to represent required reaction diversity.

### $K_4$ (Prebiotic/biological membrane continua).

- $\Theta_{\text{grad}}$ : osmotic pressure limits,
- $\Theta_{\text{perm}}$ : permeability thresholds,
- $\Theta_{\text{mem}}$ : membrane rupture and curvature thresholds.

### $K_5$ (Excitable continua).

- $\Theta_{\text{crit}}$ : spike initiation thresholds,
- $\Theta_{\text{stab}}$ : refractory conditions and excitability bounds,
- $\Theta_{\text{death}}$ : loss of excitability and breakdown of excitable cycles.

### $K_6$ (Cognitive continua).

- $\Theta_{\text{expr}}$ : limits of binding capacity and representational complexity,
- $\Theta_{\text{crit}}$ : prediction–error thresholds,
- $\Theta_{\text{stab}}$ : memory coherence and model–stability limits.

### $K_7$ – $K_8$ (Social and civilizational continua).

- $\Theta_{\text{exist}}$ : thresholds of trust and legitimacy,
- $\Theta_{\text{expr}}$ : institutional capacity and expressive limits,
- $\Theta_{\text{death}}$ : systemic collapse and breakdown of institutional and infrastructural cycles.

### $K_9$ – $K_{10}$ (Theoretical and meta–theoretical continua).

- $\Theta_{\text{expr}}$ : expressive capacity of theories and meta–languages,
- $\Theta_{\text{dim}}$ : thresholds for meta–level expansion,
- $\Theta_{\text{death}}$ : inconsistency, incoherence or structural incompatibility in the space of models.

## 4.8 Summary

The extended threshold landscape provides the structural backbone for all qualitative change in OC. Thresholds:

- carve out  $\Omega(K)$  and define  $\partial\Omega(K)$ ,
- govern phase transitions, dimensional emergence and collapse,
- mediate cascades that connect gradients, permeability and structural failure,
- encode expressive and embedding limitations that can kill continua even without energetic overload.

Together with the boundary and patch geometry of Section 3, the threshold landscape completes the description of how continua are born, maintained and destroyed in the OC framework.

## 5 Module Architecture and Cross-Level Structures

This section provides a high-level overview of the extended structural modules of the Ontology of Continua. Each module is developed in the technical atlas; here we summarise their roles and relations so the main spine has a complete bridge between formal doctrine and the Core 1.3 closure chapters.

The extended modules are:

- **K-levels master** (Section ??): global overview of all continua  $K_0$ – $K_{12}$ , their state spaces, axes, thresholds and core processes.
- **Embedding spaces / M-spaces** (Section ??): definition of meta-spaces  $M_x$  that constrain and parameterise the evolution of continua  $K_x$ .
- **Cross-level / cross-K structures** (Section ??): operators, mappings and landscape structures that link different levels of the hierarchy and generate phase transitions between them.
- **Processes** (Section ??): unified description of evolution operators  $E = (F, G, H, Q, R, S, U)$  across levels, including tension dynamics, thresholds, flows and cycles.
- **Cycles** (Section ??): structural and dynamical cycles  $C_j$  as carriers of continuity, memory and temporal organisation.

- **Predictions** (Section ??): level-specific and cross-level predictions, empirical signatures and qualitative regimes implied by the formalism.
- **Experiments** (Section ??): conceptual and concrete experimental designs to probe continua, transitions and threshold landscapes.
- **Falsifiability structure** (Section ??): explicit conditions under which the Ontology of Continua can be refuted, constrained or forced to restructure its core.
- **Jets / local evolution structure** (Section ??): local expansions of the evolution operator and short-time dynamics around given configurations of  $K_x$  and  $M_x$ .

Together, these modules refine and operationalise the core formalism across levels and applications, while maintaining a single coherent hierarchy of continua and meta-spaces.

## 6 Source Witnessing, Theorem Fate, and Publication Boundary

This chapter fixes the publication boundary of Core 1.3. Its task is to state which scientific authority is canonical, how source witnessing is enforced, how theorem fate is separated across outward artifacts, and how later channels must remain aligned with the source corpus, SPOT, and flagship monograph. It does not re-argue the formal doctrine or theorem spine; those burdens belong to the surrounding chapters and appendices.

### 6.1 Canonical scientific authority

The canonical scientific authority of the release is the SPOT projection `releases/oc_core_1_3/editorial/OC_CORE_1_3_SCIENCE_SPOT_latest.json` (surface id `OC_CORE_1_3_SCIENCE_SPOT`; current SHA recorded in its `metadata.repo_sha` field). The source-owned science corpus at `releases/oc_core_1_3/editorial/science_sources/` is the authoring and witness locus from which that SPOT is projected. Historical source witnesses, including earlier Core 1.2/Core 1.3 repair baselines, remain provenance anchors; they are not the current projection authority. The present master monograph is the flagship reader-facing projection and compiled reference volume derived from those records. It keeps the orientation ladder, formal doctrine, theorem route, empirical discipline, synthesis, practical utility, and audit layers visible together, but it does not displace the SPOT as the final authority on claim scope, promotion state, or proof boundary.

This rule matters because foundational programmes accumulate several different textual layers over time: source-native definitions and theorems, explanatory prose, repair notes, channel-specific summaries, and later operational packets. Core 1.3 treats those layers as distinguishable, not interchangeable.

### 6.2 Source witnessing

Core 1.3 is source-first in the strict sense that the flagship claims used in public release must remain bound to public source witnesses rather than to late summary prose alone. The point is not archival fetishism. The point is to stop drift between five things that often blur inside large research programmes:

- theorem-bearing statements,
- structural and interpretive explanation,
- repair and release notes,
- condensed public packets,
- and later operational projections.

When that boundary blurs, a summary sentence can begin to masquerade as the theorem itself, or a release dashboard can begin to replace the actual scientific record. Core 1.3 blocks that failure by keeping

source witnessing bound to the source-owned science corpus and the canonical SPOT projection rather than relegating it to hidden control files.

### 6.3 Theorem fate and scope discipline

The present release also separates theorem fate explicitly. Not every inherited row serves the same role in every outward artifact.

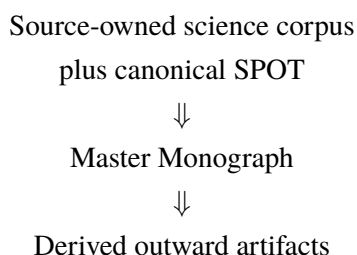
- `THEOREM_NATIVE` rows may carry positive theorem-native claims when their proof and dependency route remains exact.
- `BRIDGE_ONLY` rows may remain scientifically useful under explicit scope notes, but they cannot be promoted as theorem-native.
- `FRAME_ONLY` and `EXTENSION` rows belong in support layers, appendices, narrower companion artifacts, or future work rather than in the positive body of every outward paper.
- `REFUTED` rows are excluded from the positive proof path. They may be mentioned only as negative audit history or as a falsifier record, not as live support.
- Frontier or hypothesis material may be discussed only under visibly labelled frontier language and may not be silently promoted by summary prose.

The public positive closure claim is scoped exactly to `CORE_1_3_SCIENCE_ONLY`. `FRAME_ONLY` rows may still close as frame-level PASS rows inside the synthesis surface, but they do not become standalone theorem-native empirical promotions. `EXTENSION` and `REFUTED` rows remain outside the public PASS scope for different reasons: `EXTENSION` rows are frontier/support lanes awaiting later promotion or rejection, while `REFUTED` rows are negative-audit lanes. Neither class can carry positive closure claims in this release. For avoidance of doubt, a `FRAME_ONLY` PASS row is excluded from the public positive closure claim unless a later release promotes it into `CORE_1_3_SCIENCE_ONLY` with the required theorem-native support.

This is not retreat. It is a condition of honest publication. The monograph can remain large without rhetorical inflation only if the reader can tell which rows are theorem-bearing, which are bounded, and which are support or frontier material.

### 6.4 Relation to journal-core and companion artifacts

Core 1.3 therefore adopts a one-way reader-facing publication logic:



This diagram describes publication order, not the full provenance graph of the release package. Several editorial JSON surfaces, generated science chapters, and release matrices are direct SPOT projections or mirrors of other governing canonical surfaces. They do not become true merely because the master monograph cites them, and they are not descendants of the monograph alone. The monograph is the flagship reading object; the source-owned science corpus remains the authoring and witness locus, and the SPOT remains the canonical projection authority for release surfaces.

The journal-core article is narrower by design. Companion dossiers, matrices, and release surfaces are narrower or more operational by design. Those artifacts may compress, subset, or reorganise the flagship

material for a specific task, but they may not enlarge the scientific claim, widen notation, or outrun the theorem-fate discipline fixed here.

The practical consequence is simple. If an outward reader-facing artifact says something stronger than the source corpus and SPOT authorise, the outward artifact is wrong, even if it is shorter, cleaner, or easier to market. The flagship boundary is the publication rule that enforces that ceiling; it is not an additional source of scientific authority.

## 6.5 Downstream projection rule

The same boundary stabilizes later use in review, collaboration, product, business, and public communication. Different channels may legitimately cite different parts of the flagship volume:

- the theorem route when bounded proof is the issue,
- the empirical route when promotion discipline is the issue,
- the practical route when use and comparison are the issue,
- the audit appendices when challenge and traceability are the issue.

What they may not do is invent a second scientific authority. Source witnessing, theorem fate, and publication boundary exist so that external use can remain concise without becoming interpretively lawless.

**Bridge to the next chapter.** With the publication boundary fixed, the next task is internal consistency: which statement classes the flagship volume uses, why the hierarchy runs through  $K_{12}$ , and how upper levels remain lawful without rhetorical inflation.

## 7 Core 1.3 Foundational Consistency Dossier

This chapter fixes the internal consistency contract of the flagship volume. Its purpose is not to replace the formal chapters with editorial commentary, but to make three things readable in one place:

- why the present hierarchy is fixed as  $K_0$  through  $K_{12}$  rather than by an arbitrary cutoff;
- the full statement-class discipline for axiomatic, derived, empirical, structural, interpretive, and frontier claims;
- how the master monograph and the narrower journal-core extraction are related without silent drift in scope or notation.

### 7.1 Why the hierarchy extends through $K_{12}$

Core 1.2 already contains enough structure to motivate a release hierarchy larger than  $K_{10}$ . The point is not that higher labels are rhetorically attractive. The point is that the audited source corpus already distinguishes:

1. lower-level continua whose axes, thresholds, and flows are internal to a given model class;
2. theoretical and meta-theoretical continua that organise theories and model-spaces themselves;
3. a further stratum in which operators, meta-logics, and model-spaces become explicit objects of structural evolution;
4. and an upper bounding layer in which global coherence conditions across such higher-order structures must themselves be represented.

That is why Core 1.3 carries the public hierarchy through  $K_{12}$  rather than stopping at  $K_{10}$ . This is a release-level design rationale, not a standalone impossibility proof. The supporting route is the K-level doctrine in Section ??, the theorem-navigation discussion in Section 8, and the worked preservation test in Section 9. Those sections assign  $K_{11}$  to evolving meta-theoretical objects and  $K_{12}$  to global compatibility and integrability conditions. The present release does not claim a proof that no further extension is conceivable in principle; it claims only that  $K_{12}$  is the highest level lawfully carried by the current audited source corpus and therefore the last public level of this release.

### 7.2 Scientific role of the upper levels

**Level  $K_{11}$ .**  $K_{11}$  is the first level at which formal ontologies, operator systems, meta-logics, and model-spaces are themselves treated as structured continua subject to admissibility, transformation, and collapse conditions. This level is needed once one studies not only theories, but the evolution of the spaces in which theories and theory-building operators live.

**Level  $K_{12}$ .**  $K_{12}$  is not introduced as a mystical completion layer. It is the minimally specified upper coherence stratum required when one needs to discuss global compatibility, boundedness, and failure modes across families of  $K_{11}$ -objects. Its role is structural rather than triumphalist: it prevents the hierarchy from pretending that there is no formal problem above recursively evolving meta-theoretical spaces.

### 7.3 Statement classes

To keep the monograph readable without blurring scientific status, Core 1.3 uses the following proof-status vocabulary in this chapter. It is the local theorem and claim-scope taxonomy; Section 8 adds the downstream operational inspection burden for replay, comparator, falsifier, and escalation rules.

- **Axiomatic statements** fix primitive assumptions, admissibility conditions, or identity rules.

- **Derived statements** are theorem-bearing statements with explicit support dependence and proof obligations.
- **Empirical statements** require a standalone source-owned domain packet, canonical theorem-native trace, pinned data route, numerical parameter law, observable binding, locked held-out replay procedure, comparator audit, and explicit falsifier before promotion.
- **Structural statements** describe the organisation of the framework, the hierarchy, and the mapping between modules or levels.
- **Interpretive statements** explain what the formal structure means, which reader it is for, and what later projections may test.
- **Frontier statements** may remain in the research program, but may not enter positive outward science without formal promotion.

These local classes are not assumed to be disjoint. When a sentence could fit more than one class, its public status is assigned by semantic role and proof function rather than by surface phrasing:

1. an explicit contradiction of a promoted claim first flags a refutation-candidate or surface-contradiction case for SPOT audit;
2. empirical content is inspected by packet, data route, replay, comparator, and falsifier support before any prose label is trusted;
3. derived theorem content is inspected by its proof route and dependency graph;
4. admitted primitive assumptions remain axiomatic premises;
5. structural mapping and hierarchy claims remain structural even when they mention possible later use;
6. explanatory, audience-facing, or didactic sentences remain interpretive unless they assert theorem or empirical force;
7. unpromoted research-program proposals remain frontier statements;
8. pure structural or interpretive content keeps the SPOT-recorded `FRAME_ONLY` or `EXTENSION` class unless it is explicitly attached to evidence-bearing bridge material;
9. `BRIDGE_ONLY` is a stable evidence-support class for rows or packets that carry bridge support under explicit scope notes; it is not a theorem-native promotion queue.

Words such as “frontier”, “candidate”, or “projection” are therefore diagnostic signals, not the class assignment rule itself.

These prose classes are a local proof-status taxonomy, not a replacement for the canonical SPOT scientific-class and closure-verdict vocabulary. Release verdicts such as `PASS` and `FAIL_CLOSED` remain a separate verdict layer. The mapping is:

- Axiomatic statements are admitted premises or identity rules. They may sit inside a `THEOREM_NATIVE` route as named assumptions, but the axiom itself is not thereby converted into a proved theorem.
- Derived statements map to `THEOREM_NATIVE` when the support route is closed.
- Empirical statements map to `THEOREM_NATIVE` only after the source-owned packet, theorem-native trace, pinned data route, numerical parameter law, observable binding, locked held-out replay procedure, comparator audit, and falsifier gates pass.
- Structural and interpretive statements map to `FRAME_ONLY` or `EXTENSION` as recorded by SPOT unless the statement is attached to an evidence-bearing bridge row.

- Frontier statements map to `EXTENSION` until a later section or release promotes them through the same theorem-native or empirical gates used for other positive claims, or records an explicit refutation.
- Fail-closed bridge, frame, or extension lanes retain the class recorded by SPOT; closure failure by itself is not refutation.
- Statements map to `REFUTED` only when SPOT records an explicit refutation rather than a blocked promotion.

This chapter classifies the statement classes that affect theorem status, structural scope, interpretation, empirical promotion, and frontier boundaries. Operational route discipline is handled later in the theorem-roadmap and proof-machinery chapters. Editorial-bridge language is not a seventh proof class; it is an interpretive and release-consistency label for statements that connect the monograph, journal core, dossiers, and public package without adding theorem force. Operational statements are the later exact-match, held-out replay, statistical-threshold, tail-breach, comparator, falsifier, and escalation rules attached to empirical promotion. Structural statements are reviewed by scope, dependency, and source-boundary checks instead. Both kinds of control are handled after the statement class, numerical parameter law, and observable binding have already been fixed here, and neither becomes a seventh proof-status class.

The journal-core extraction is allowed to narrow this body, but not to blur these classes into one another. An interpretive sentence may explain a theorem; it may not replace the theorem. A structural remark may motivate a projection; it may not silently serve as proof of that projection.

## 7.4 Proof-bearing and nonclaim layers

The master monograph therefore separates three layers explicitly:

1. the inherited formal backbone that still bears the structure of the core theory;
2. the Core 1.3 repair layer that clarifies witness discipline, theorem fate, chronology, and publication boundary;
3. the bounded nonclaim layer that states what the present release does not yet defend, even when the broader research program still pursues it.

This separation matters for hostile review. A reviewer should be able to ask, for any sentence that sounds strong: is it axiomatic, derived, empirical, structural, interpretive, or frontier? If it is derived, what supports it? If it is empirical, what packet, data route, replay, comparator, and falsifier support it? If it is structural or interpretive, what exactly is it not claiming? If it is frontier, where is it visibly barred from positive release authority?

## 7.5 Relation between master monograph and journal core

The canonical SPOT is the scientific authority of the release, and the source-owned science corpus is the authoring and witness locus behind that projection. The master monograph is the flagship reader-facing reference because it carries the full theory, the reader ladder, the appendices, and the explicit publication boundary in one place. The *Journal Core* is the bounded article artifact. The *journal-core extraction* is the operation that selects one defensible theorem-bearing route for editorial inspection without pretending that the whole monograph is already compressed into that shorter artifact.

For that reason the extraction rule is one-way:

Master Monograph  $\longrightarrow$  Journal Core,

with explicit preservation of notation, nonclaim boundaries, and theorem scope. The extraction procedure may omit detail. The Journal Core may not enlarge the claim. The public closure boundary remains the `CORE_1_3_SCIENCE_ONLY` scope stated in Section 6. `FRAME_ONLY` closures may be `PASS` as



frame support, but they do not become standalone empirical promotions. `BRIDGE_ONLY` rows are scope-bounded evidence support rather than public promoted closure; they are not re-labelled as theorem-native `PASS` rows unless a separate source-owned theorem-native packet is created and passes the canonical gates. `EXTENSION` and `REFUTED` lanes remain outside public `PASS` authority.

## 7.6 Reader-facing consequence

This chapter exists because scientific rigor and readability do not pull in opposite directions here. The more explicit the dossier becomes about hierarchy, statement class, and proof boundary, the easier it becomes for three readers at once:

- the editor who needs a clean overview of what is actually defended;
- the formal reader who needs exact scope and dependency discipline;
- the non-specialist scientific reader who needs a coherent narrative entry point before diving into the full technical shell.

**Bridge to the next chapter.** With the statement classes, hierarchy boundary, and master-to-derivative relation fixed, the next chapter narrows further to one task only: the shortest lawful proof-navigation route through the load-bearing theorem spine.

## 8 Theorem Roadmap and Proof Navigation

This chapter is the clearest compact map of the proof-bearing route. Its task is to show where the load-bearing theorem spine lies, which statement classes carry which scientific burden, and where a hostile reviewer should probe first. It does not replace the formal chapters or the reader guide. The reader guide owns entry routes; this chapter owns proof navigation.

### 8.1 Load-bearing theorem spine

At the level of the bounded journal-core claim, the present release depends on a relatively small load-bearing spine even though the monograph itself remains much larger. The schematic route below is compact rather than proof-theoretic minimality in the strict sense: it keeps the residue and rebirth branch visible because the defended article-level theorem classifies post-collapse identity, not death alone. The spine can therefore be stated schematically as dependency notation rather than object-language implication. To avoid making the route look like a proof calculus, the support map is written as labelled review edges:

1. **Edge 1.** Axiom 3.2 together with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3, and Source Theorem 3 supports Lemma 1.
2. **Edge 2.** Axiom 3.3 together with Definition 12.6 supports Source Corollary 3.2, and Source Corollary 3.2 supports Lemma 3.
3. **Edge 3.** Lemma 1 together with Definition 12.5 supports Lemma 2.
4. **Edge 4.** Lemma 2 is the sole premise recorded for Theorem A in the canonical proof-obligation registry.

Definitions in this list are prerequisite nodes, not independent theorem premises. The word “supports” marks an admitted release support step rather than a primitive implication operator of OC.

Lemma 3 remains an admitted rebirth-control obligation in the surrounding roadmap, but it is not a premise of Theorem A in the canonical proof-obligation registry. Its load-bearing role is different: it supports the companion rebirth-control statement, Proposition B. Whenever a compressed article proof discusses the death–residue–rebirth package in one paragraph, Lemma 3 should be read as supporting Proposition B, while Theorem A itself remains the death-and-residue classification routed through Lemma 2.

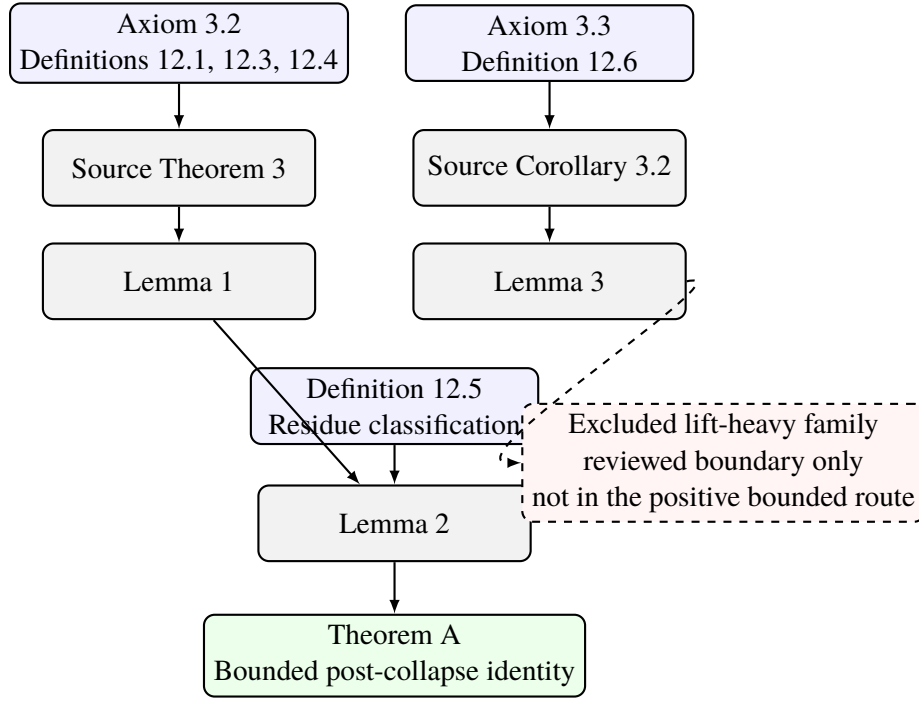
**Proposition B (article-local rebirth control).** Within the article-level extraction, after  $K$  has lawfully collapsed so that  $\Omega(K) = \emptyset$  and  $k(K, t) = 0$ , a later object  $K'$  is classified as rebirth if and only if  $K'$  is live, is numerically distinct from the dead original continuum  $K$ , appears either in the same-embedding branch or in an explicitly witnessed related-embedding branch, inherits at least one declared residue support element, and satisfies the relevant birth conditions. The identity rule supplies the reason this non-identity condition is forced rather than optional. This proposition is a companion control statement for the rebirth branch; it is not a premise of Theorem A.

In this roadmap, the *Source* prefix marks a witness projection from the audited public LaTeX source corpus and archival compiled PDF described in the source-audit appendix. Such rows are carried into Core 1.3 as source-witnessed support nodes and are inspected through the canonical proof-obligation registry; they are not newly asserted primitive theorems of the article-level extraction. The unprefixes lemmas, Theorem A, and Proposition B are the article-level derived rows assembled from that witnessed support. Theorem A uses the death-and-residue spine; Proposition B uses the rebirth-control branch.

Axioms, source theorems and corollaries, lemmas, and Theorem A carry proof-force nodes; definitions fix the terms used by those nodes.

The point of making that chain explicit is not to pretend that the whole framework is reducible to one diagram. The point is to show that the bounded claim has an auditable positive core rather than a cloud

of motivational prose. Figure 1 also marks one excluded lift-heavy temptation route as a hostile-review boundary; it is reviewed precisely so that it cannot be mistaken for admitted support in the positive chain. The Corollary 3.2 / Lemma 3 branch is shown as an admitted rebirth-control branch for Proposition B beside the theorem route rather than as an antecedent of Theorem A. In this roadmap the related-embedding branch is represented by one auditable article-local witness, not by an exhaustive list of every admissible source-level related-embedding witness. This is the same branch denoted  $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$  in Section 9: a named structure-preserving map  $h : \mathcal{M} \rightarrow \mathcal{M}'$ , a residue element  $r \in R_{\text{res}}(K)$ , admissible-region membership  $h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}')$ , and preservation of the threshold and admissibility predicates on the residue-bearing substructure. Section 9 and the journal core state that witness test explicitly.



Solid arrows: admitted theorem route and admitted rebirth-control branch  
Dashed arrows: excluded temptation route reviewed but not admitted

Figure 1: Load-bearing theorem spine of the bounded Core 1.3 claim. Solid arrows mark admitted support. The positive Theorem A route terminates at Lemma 2; the Lemma 3 branch remains visible as admitted rebirth-control support beside that route. The dashed branch marks an excluded lift-heavy route that is reviewed as a boundary condition rather than admitted into the positive proof. The larger monograph remains necessary because the surrounding chapters fix the ontology, operator shell, K-level doctrine, collapse grammar, empirical bar, and audit layers that keep this spine scientifically interpretable and externally citable.

## 8.2 What surrounds the spine

The theorem spine has a small proof dependency, but it is interpreted and audited inside the wider manuscript in five distinct ways:

1. the formal doctrine chapters fix admissibility, thresholds, operators, levels, branching, and falsifiability;
2. the consistency and source-boundary chapters fix statement class, theorem fate, and publication scope;

3. the proof-machinery chapter and appendix fix revision, comparator, compression, and proof-obligation discipline;
4. the operational and empirical chapters fix promotion, replay, and falsifier discipline;
5. the appendices keep the audit and atlas burden inside the same volume.

That is why the monograph is allowed to be larger than the defended article without becoming rhetorically loose. These surrounding chapters are publication and audit scaffolding, not additional premises of Theorem A.

### 8.3 Statement classes and burden

The table below is a proof-navigation map for reviewer use. It follows the statement-class taxonomy of Section 7. The proposition row is a navigation-only subdivision of derived statements, not a new proof-status class. Operational rules remain a downstream inspection burden rather than a separate proof-status class.

| Statement class             | Scientific burden                                                                                                                                                                              | Where to inspect it                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Axiomatic statements        | Fix primitive admissibility, identity, or threshold conditions.                                                                                                                                | Axioms Appendix, Notation Appendix, and Sections 3–??.                             |
| Derived theorems and lemmas | Carry the proof-bearing positive body.                                                                                                                                                         | Sections 2, ??, 8, Reviewer Navigation Matrix, and Proof Machinery Appendix.       |
| Article-level propositions  | Carry companion control statements that are derived from the same audited support but are not premises of Theorem A. Proposition B is the rebirth-control example.                             | Sections 8, 9, and Journal Core Bridge Appendix.                                   |
| Structural statements       | Explain hierarchy, module layout, and relation between levels or artifacts.                                                                                                                    | Sections ??–??, Section 5, Section 6, and Section 7.                               |
| Interpretive statements     | Explain meaning, nonclaims, and downstream boundaries.                                                                                                                                         | Sections ??, 9, ??, Source Audit Appendix, and Reviewer Navigation Matrix.         |
| Empirical statements        | Require a source-owned domain packet, theorem-native trace, numerical parameter law, observable binding, pinned data route, held-out replay, comparator audit, and falsifier before promotion. | Sections ??–?? and Appendices ??–??.                                               |
| Frontier statements         | Mark lawful future work without positive release authority.                                                                                                                                    | Section ??, Appendix ??, and the frontier rows in the source-owned science corpus. |

Table 2: Statement classes and their inspection burden in Core 1.3. The table does not replace the formal chapters; it simply prevents proof, interpretation, empirical promotion, and frontier language from quietly changing roles. Replay, comparator, falsifier, and escalation rules are downstream control obligations attached to those classes.

All inspection targets in Table 2 are label-checked against the English master source. If a release wrapper changes one of those labels, the table must be regenerated or patched before the roadmap is treated as a mechanically followable audit route.

### 8.4 Shortest hostile-review checklist

If a reviewer wants the most efficient attack sequence, the logically hardest checks remain the following:

1. verify that collapse is tied to loss of admissible realization rather than to loose metaphor;

2. verify that the irreversibility rule forbids restoration of the original continuum as numerically the same entity;
3. verify that residue and rebirth are controlled classifications rather than narrative labels;
4. verify that no excluded lift-heavy theorem family is smuggled back into the positive bounded route;
5. verify that the journal-core extraction says nothing stronger than the source-owned science corpus and canonical SPOT permit; the flagship monograph is then checked only as a derived comparison surface;
6. verify that the K-level doctrine, especially  $K_{11}$  and  $K_{12}$ , is structurally justified rather than rhetorically inflated.

If the release fails there, later prose cannot rescue it. If it survives there, a slower full reading becomes worth the effort.

**Bridge to the next chapter.** With the proof route mapped, the next chapter changes mode deliberately. It no longer navigates the spine abstractly; it shows how the formal grammar should be read through worked examples, counterexamples, and bounded interpretive checks.

## 9 Worked Examples, Counterexamples, and Reader-Oriented Interpretive Checks

This chapter turns the formal grammar into bounded reading moves. Its purpose is to show what the theorem route licenses, what it forbids, and what would count as a real counterexample at the release boundary. It does not replace the proof route with storytelling; it tests whether the reader can apply the proof route without hand-waving. The examples therefore show how the formal grammar should be read, why the K-level doctrine is not an arbitrary count, and how the master volume disciplines the shorter journal-core extraction. In these examples, “declared” means named inside the local example data or in a cited support route before the classification or reduction test is applied; it never means supplied afterward to rescue a desired reading.

### 9.1 Worked Example 1: collapse, residue, and rebirth in a minimal admissible-state model

Consider a continuum  $K$  whose admissible realization at time  $t$  is described by

$$\Omega(K, t) = \{x \in [0, 1] \mid \phi(x, t) \leq \vartheta(t)\},$$

where  $\phi(x, t)$  is an incompatibility load and  $\vartheta(t)$  is the currently lawful scalar threshold. Suppose that at an earlier stage  $t_0$  the set is nonempty, for example because  $\phi(x, t_0)$  is below threshold on a compact interval  $[a, b] \subseteq [0, 1]$ .

Now assume that the embedding regime changes, the threshold tightens, or the internal evolution operator drives the incompatibility load upward so that the declared collapse time is  $t_1 = t^*$ , the approaching path satisfies the one-sided source condition  $\lim_{t \nearrow t^*} k(K, t) = 0$ , and at  $t_1$ ,

$$\forall x \in [0, 1] \quad \phi(x, t_1) > \vartheta(t_1).$$

Then  $\Omega(K, t_1) = \emptyset$ . In the language of the source-native death condition, this full witness package is not merely a bad regime or a temporarily unproductive phase. Because the pointwise admissibility indicator  $\chi_\Omega(s_K(t_1); K, t_1)$  is then zero, the continuumness measure also collapses to  $k(K, t_1) = 0$  under the unified Core 1.3 definition. The implication is immediate because  $k$  is defined as the product whose first factor is the pointwise admissibility indicator. It is therefore the lawful death of the continuum under the current admissibility regime.

The important interpretive point is what follows next. Suppose that after collapse, the embedding space still preserves some structured trace: constraints, marks, gradients, memory residues, or other inherited structure that a later continuum can exploit. That trace is not the original live continuum, because the admissible realization has already become empty. It is residue: a surviving structural aftermath without live identity.

Finally suppose that at time  $t_2 > t_1$ , after changes in thresholds, embedding, or generative conditions, a new nonempty admissible realization appears for some later object  $K'$ . The lawful classification is not that the original continuum has revived. The irreversibility rule forbids that reading. The correct classification is rebirth only when the full predicate of Definition 12.6 is met: the later live continuum appears in the same or a related embedding space, inherits at least one structural element from residue, and satisfies the relevant birth conditions. The separate identity rule in Section ?? then says why that later live continuum is numerically new rather than a restoration of the original. If a route uses the related-embedding option rather than the same embedding space, the relation must be tested against the residue and threshold clauses rather than inferred after the fact to save a desired rebirth reading. In this article-level use, one auditable compatibility witness for “related” is the explicitly scoped condition. Let  $\Omega_{\text{amb}}(\mathcal{M}')$  denote the ambient admissible region declared for the proposed embedding space, and let the quantified  $K'$  range over declared later-candidate continua in  $\mathcal{M}'$ :

$$\begin{aligned} \exists K' \exists h : \mathcal{M} \rightarrow \mathcal{M}' \exists r \in R_{\text{res}}(K) \\ h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}') \text{ and } \text{Pres}_{\Theta, \Omega}(h, r; K, K'). \end{aligned}$$

For this worked example,  $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$  means exactly that the displayed line is shorthand for the full witness package described below. That package includes the residue-bearing substructure, the active residue and target index families, the index bijection, and the induced transport operator; without those data the displayed formula is not a standalone formal predicate. Here  $K'$  is the declared later candidate tested for live admissibility,  $\mathcal{M}$  is the embedding space of the collapsed continuum,  $\mathcal{M}'$  is the embedding space proposed for the later continuum,  $R_{\text{res}}(K)$  is the residue set of the collapsed continuum, and  $\mathcal{R}_K(r) \subseteq R_{\text{res}}(K)$  denotes the declared residue-bearing substructure containing  $r$ : a local residue subobject named in the witness data on which the transported threshold and admissibility data are declared. If no such substructure is declared, the related embedding branch is not available in this example.  $\text{Pres}_{\Theta, \Omega}$  means that the named map  $h$  preserves  $\mathcal{R}_K(r)$  and the threshold and admissibility predicates used to test the later live continuum. This witness is not by itself the full rebirth predicate; it supplies the related-embedding clause inside that predicate. This is the example-specific relation denoted by  $\mathcal{M} \sim_{\text{res}} \mathcal{M}'$ . The formal rebirth predicate remains Definition 12.6; the source predicate may admit other properly declared related-embedding witnesses; this worked example uses the stricter  $\sim_{\text{res}}$  witness so the related branch is auditable rather than assumed. The witness data also include the active residue-threshold index family  $I_K^{\text{res}}$ , the corresponding active target subfamily  $I_{K'}^h \subseteq I_{K'}$ , and a declared bijection  $\eta_h : I_K^{\text{res}} \xrightarrow{\sim} I_{K'}^h$ . The induced pushforward  $h_*$  applies  $h$  to the residue state and transports each named residual threshold by  $\eta_h$ . For this example,  $\text{Pres}_{\Theta, \Omega}$  is the following declared predicate:

$$\begin{aligned} &\text{Pres}_{\Theta, \Omega}(h, r; K, K') \text{ holds exactly when} \\ &h \text{ preserves } \mathcal{R}_K(r), \\ &h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(\mathcal{M}'), \\ &\mathcal{D}_{\Theta, \Omega}^h(K'; h(r)) = h_* \mathcal{D}_{\Theta, \Omega}^{\text{res}}(K; r). \end{aligned}$$

Here  $\mathcal{D}_{\Theta, \Omega}$  denotes the local admissibility data: the pointwise admissible-membership indicator together with the active residual-threshold signs and saturated constraints. The term  $\mathcal{D}_{\Theta, \Omega}^{\text{res}}(K; r)$  is the same data restricted to the residue-index family  $I_K^{\text{res}}$ , while  $\mathcal{D}_{\Theta, \Omega}^h(K'; h(r))$  is the target-side record restricted to  $I_{K'}^h$ . More explicitly, for a finite active threshold subfamily  $J \subseteq I_Y$  of  $\Theta_Y = \{f_i^Y\}_{i \in I_Y}$  at a tested object  $Y$ , point  $y$ , and test time  $\tau$ , the restricted typed record is

$$\mathcal{D}_{\Theta, \Omega}^J(Y; y) = (\chi_{\Omega}(y; Y, \tau), (\text{sgn } f_i^Y(y))_{i \in J}, \{i \in J : f_i^Y(y) = 0\}).$$

The unreduced symbol  $\mathcal{D}_{\Theta, \Omega}(Y; y)$  abbreviates the special case  $J = I_Y$ ; it is not the record compared in the related-embedding witness. Here  $I_K^{\text{res}}$  is the subfamily of active thresholds declared on the residue-bearing substructure, and  $I_{K'}^h$  is the corresponding active target subfamily used to test  $\Omega(K')$  along this witness. The transport satisfies  $f_{\eta_h(i)}^{K'} \circ h = f_i^K$  on  $\mathcal{R}_K(r)$ . Consequently the saturated set is transported as

$$\eta_h(\{i \in I_K^{\text{res}} : f_i^K(r) = 0\}) = \{j \in I_{K'}^h : f_j^{K'}(h(r)) = 0\}.$$

The equality therefore says that the named map transports the residue-bearing threshold and admissibility data used to test the later live continuum; it is not an informal resemblance clause. The pushforward  $h_*$  is a witness-local transport-plus-target-evaluation operator rather than a source-only pushforward: it applies  $h$  to the residue state, carries each named residual threshold  $f_i$  to the corresponding declared residual threshold  $f'_{\eta_h(i)}$ , preserves the sign relation  $\leq 0$ , preserves the transported saturated subset, and evaluates the target membership component by the separate predicate  $h(r) \in \Omega(K')$ .

This example is deliberately modest. It does not claim that every physically, biologically, or socially interesting regime can already be reduced to such a scalar picture. What it does show is how the theorem grammar should be read once the support chain is accepted. A hostile reviewer may reject the premises or the admissibility logic. What the reviewer should not be allowed to say is that the release leaves the post-collapse identity consequences undefined.

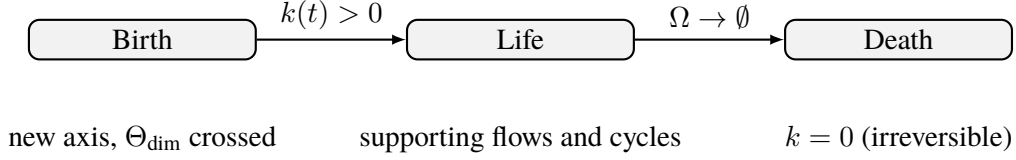


Figure 2: Minimal life-cycle schematic for the bounded collapse grammar. The point of the figure is not to offer a domain-specific simulation but to show the exact interpretive burden of the formal rules: in this minimal model, the object remains live only while the admissible region is nonempty, the required cycles remain present, and  $k(K, t) > 0$ , with supporting flows not dominated by destructive flows on the relevant cycles. Admissible-state collapse kills it, and any later live continuum is classified as rebirth rather than persistence only when the residue, embedding-space, and birth-condition clauses of Definition 12.6 are satisfied and the identity rule rules out restoration of the dead original.

## 9.2 Worked Example 2: why $K_{11}$ and $K_{12}$ are not decorative

One of the easiest ways to make a foundational framework look more sophisticated than it is would be to keep extending its hierarchy labels without necessity. Core 1.3 blocks that temptation by forcing the reader to ask what would break if the hierarchy stopped at  $K_{10}$ .

At the level of  $K_9$ , the theory already treats theories, paradigms, and ontological schemes as structured objects. At  $K_{10}$ , it treats model-of-models recursion and self-reference as part of the formal landscape. That still leaves two scientific problems open.

1. How do operator families, meta-logics, and model-spaces themselves evolve as structured continua rather than as static meta-language?
2. How are global coherence conditions expressed once multiple such higher-order objects coexist and partially constrain one another?

Stopping at  $K_{10}$  would force those problems back into informal commentary. The public source corpus, however, already contains the beginnings of explicit treatment for both. That is why  $K_{11}$  and  $K_{12}$  are retained: not because bigger counts sound more impressive, but because the framework has already crossed the line at which the evolution of theory-space and the coherence of higher-order structures have to be represented formally.

The reader-oriented reduction check here is explicit and bounded. It is an audit heuristic for seeing what a  $K_{10}$ -only reduction would have to preserve, not a replacement for the source-owned irreducibility dossier. In this paragraph,  $\text{Car}(K_n)$  denotes the carrier set of objects assigned to level  $K_n$ , not the level label itself. Let  $\mathcal{U}_{11,12} = \text{Car}(K_{11}) \sqcup \text{Car}(K_{12})$  denote the disjoint union of objects assigned to the two tested upper-level roles. A  $K_{10}$ -only reduction is a tuple  $(\rho, \equiv_\rho, \rho_*, \lambda)$  whenever time-sensitive requirements are present, where  $\rho : \mathcal{U}_{11,12} \rightarrow \text{Car}(K_{10})$ ,  $\equiv_\rho \subseteq \mathcal{U}_{11,12} \times \mathcal{U}_{11,12}$  is the declared compatibility relation induced by the translation, and  $\rho_*$  is shorthand for the family of transports  $\{\rho_{*,X}\}_X$  that sends upper-level requirements to their proposed  $K_{10}$  counterparts. The component  $\lambda$  is the family of order-preserving interval transports  $\{\lambda_X : I_X \rightarrow I_{\rho(X)}\}_X$ . If no time-sensitive requirement is being tested,  $\lambda$  is the empty auxiliary family. Despite the glyph,  $\equiv_\rho$  is not assumed to be an equivalence relation in this test; it is only a typed compatibility predicate unless reflexivity, symmetry, and transitivity are supplied separately. For each object  $X$ , let  $\mathcal{B}(X) \subseteq \mathcal{L}_{\text{OC}}(X)$  be a finite typed set of local OC requirements, and let  $\mathcal{B}^\wedge(X)$  be its finite conjunction-closure. The allowed atom constructors are *Adm*, *Cont*, *Op*, *Id*, and *Fals*, respectively standing for admissibility, positive continuumness, operator-obligation, identity-preservation, and falsifier requirements. The notation  $\mathcal{L}_{\text{OC}}(X)$  names the finitely parameterised requirement language generated by those atom types for  $X$ , while the tested set  $\mathcal{B}(X)$  remains finite. The notation  $\mathcal{L}_{10}(\rho(X))$  names the corresponding proposed  $K_{10}$  target language. The transport  $\rho_{*,X} : \mathcal{L}_{\text{OC}}(X) \rightarrow \mathcal{L}_{10}(\rho(X))$  maps each typed atom to its proposed  $K_{10}$ -language counterpart and extends to finite conjunctions by preserving their typed clauses. For the reduction test, the satisfaction



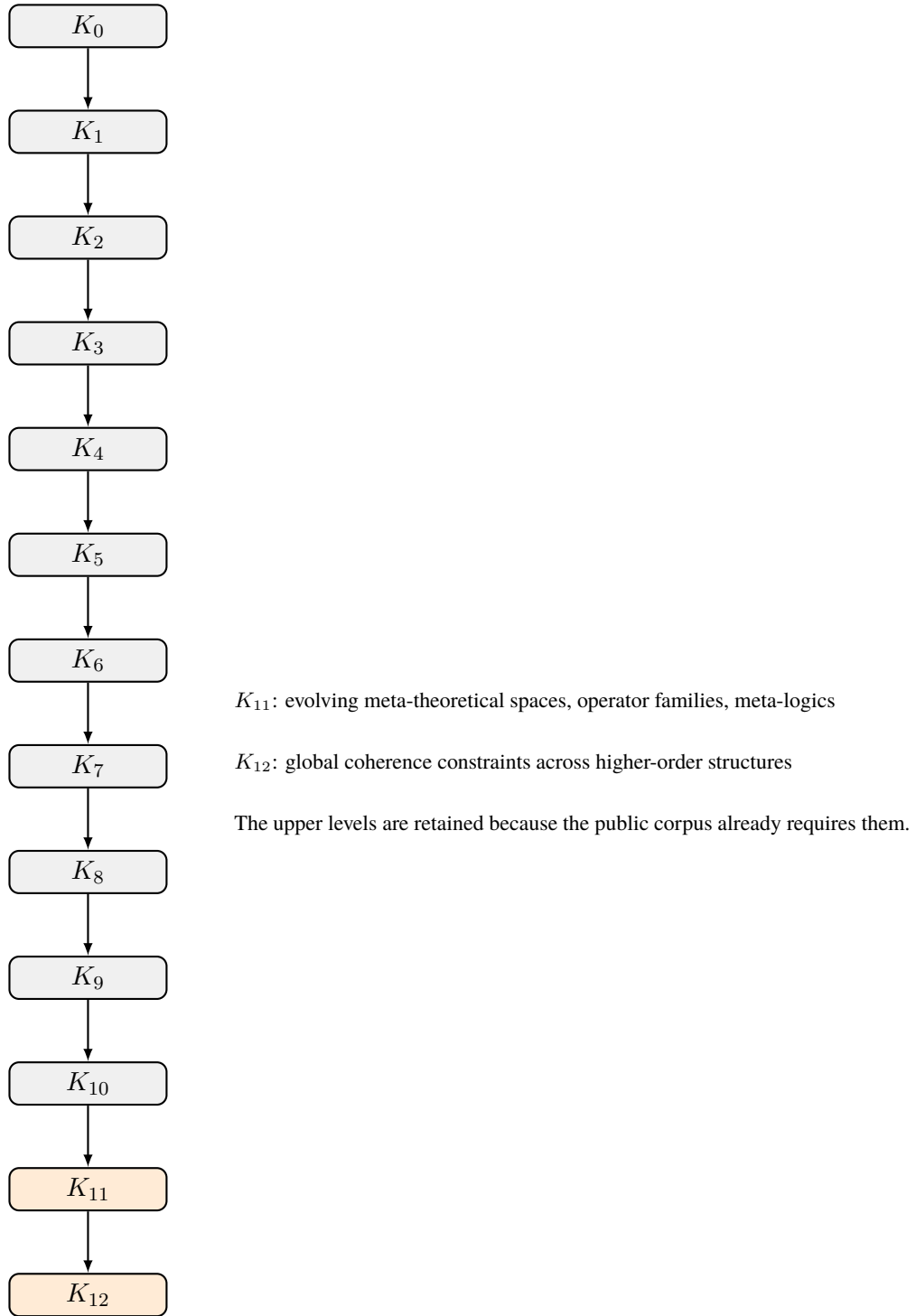


Figure 3: Extended vertical hierarchy through  $K_{12}$ . The added upper levels are justified by the need to represent evolving meta-theoretical spaces and their global coherence conditions rather than by decorative level inflation.

clauses are the following bounded clauses, where  $I_X$  is the declared nonempty finite test interval for  $X$  and  $\varepsilon \in (0, 1)$ . Let  $\text{Guard}_X(O, S, T)$ ,  $\text{IdPres}_X(\iota)$ , and  $\text{Kill}_X(\varphi)$  denote, respectively, the route-local operator-admissibility guard, identity-preservation predicate, and active falsifier predicate recorded in the proof-obligation sheet for the tested claim:

$$\begin{aligned} X \models \text{Adm}(\omega) &\iff \omega \in \Omega(X), \\ X \models \text{Cont}(\varepsilon) &\iff \forall t \in I_X, k(X, t) > \varepsilon, \\ X \models \text{Op}(O : S \rightarrow T) &\iff S \subseteq \text{Dom}_X(O), O(S) \subseteq T, \text{ and } \text{Guard}_X(O, S, T) = 1, \\ X \models \text{Id}(\iota) &\iff \text{IdPres}_X(\iota) = 1, \\ X \models \text{Fals}(\varphi) &\iff \text{Kill}_X(\varphi) = 0. \end{aligned}$$

Finite conjunctions are satisfied exactly when each typed clause in the conjunction is satisfied. Here  $X \models r$  means satisfaction of that typed requirement under the admissibility, continuumness, operator-obligation, identity, and falsifier clauses of the surrounding proof route. An operator-obligation atom is satisfied only when the relevant operator is defined on the declared source and target objects and the mapped obligation preserves the operator's stated domain, codomain, and admissibility guard; otherwise it fails. The expression  $\rho(X) \models \rho_{*,X}(r)$  means satisfaction of the translated requirement in the target language, using  $I_{\rho(X)} = \lambda_X(I_X)$  for time-sensitive atoms. The relation  $\equiv_\rho$  is a typed compatibility relation on tested objects; if it is used as a quotient, reflexivity, symmetry, and transitivity must be supplied explicitly. In this section it is used only as a burden-preservation compatibility test, not as a decorative quotient. The reduction is lossless only if

$$\forall X \in \mathcal{U}_{11,12} \forall r \in \mathcal{B}^\wedge(X) : \quad X \models r \iff \rho(X) \models \rho_{*,X}(r),$$

and only if  $X \equiv_\rho Y$  never declares compatible two objects whose admissible-state, continuumness, operator-obligation, identity, or falsifier burdens differ. This makes the compatibility relation a preservation constraint rather than a decorative quotient. In the reduction test, source-object compatibility and source-to-target preservation are separate checks. The relation  $\equiv_\rho$  constrains compatibility among source objects only; the pair  $X, \rho(X)$  preserves identity only when the admissible-state, continuumness, operator-obligation, identity-preservation, and falsifier burdens of  $X$  match their translated  $K_{10}$  counterparts under  $\rho_{*,X}$ . Residue language is only an analogy here, not the identity criterion for reduction. The displayed condition is the local, formally quantified preservation test. It is necessary for any reduction claim, but this worked example also records a separate non-definitional coherence audit. For a finite family of tested objects  $F \subseteq \mathcal{U}_{11,12}$ , chosen requirements  $r_X \in \mathcal{B}^\wedge(X)$ , and a declared common meta-language  $\mathcal{L}_F$ , the audit must also declare translations  $\iota_{F,X} : \mathcal{B}^\wedge(X) \rightarrow \mathcal{L}_F$ . The family-level test is the finite conjunction  $\bigwedge_{X \in F} \iota_{F,X}(r_X)$  inside  $\mathcal{L}_F$ , together with any cross-object coherence clauses asserted for that family. In this heuristic check, a  $K_{10}$ -only reduction is flagged if it preserves every object locally while losing a declared cross-framework coherence obligation carried by  $K_{12}$ . Thus, in the audit sense used here, “reducible without loss” would require preserving inside  $K_{10}$  alone all three burdens that the release assigns to the upper levels:

1. explicit evolution of meta-theoretical objects as lawful structured states,
2. a distinction between local meta-model recursion and global cross-framework coherence conditions,
3. and the admissibility or collapse conditions attached to those higher-order objects.

Accordingly, a critic would need to provide one explicit  $K_{10}$ -only reduction map that preserves those burdens simultaneously. If such a map exists, the upper levels should be demoted. If, however, the framework genuinely needs one stratum for evolving meta-theoretical objects and another for cross-framework coherence conditions, then the upper levels are not optional. Core 1.3 takes the second position and, crucially, states the preservation test under which that position could fail. The role split is explicit:  $K_{11}$

carries the evolution of meta-theoretical objects and operator/model spaces, while  $K_{12}$  carries global cross-framework coherence, boundedness, and integrability conditions. The same preservation criterion is used below whenever the table speaks of a later continuum being numerically identical to the dead original. In this section that means preservation of original live identity: the original admissible-state and continuumness conditions must remain live, not merely that a later object can be related to residue.

### 9.3 Interpretive Check 3: how the journal core is lawfully extracted from the monograph

Readers often distrust large monographs because they have seen too many cases in which a short article quietly says something stronger than the full technical work behind it. Core 1.3 is designed to make that drift visible and prevent it.

The lawful extraction rule is asymmetric:

Master Monograph  $\longrightarrow$  Journal Core.

The journal core may compress, but it may not enlarge. That principle has several operational consequences.

1. The journal core can suppress much of the didactic shell, but it must preserve the exact positive theorem scope.
2. It can omit broad projection discussions, but it may not promote any projection into accomplished proof.
3. It can shorten the literature screen, but it may not falsify the priority or comparator boundary set in the master.
4. It can move quickly through notation once the master has fixed it, but it may not invent a second notation universe.

This is not an additional scientific premise. It is a release-governance rule for packaging and review. The monograph is where the broader framework is taught, bounded, and audited. The journal core is where one narrower result is exposed for fast hostile inspection. If the two diverge, the release should be treated as broken.

### 9.4 What would count as a real counterexample

To keep the examples falsifiable, it is useful to state what kind of observation or formal result would genuinely pressure the present release.

### 9.5 Why examples belong in a strict scientific monograph

Examples are often treated as a concession to weaker readers. In foundational work that is a mistake. A disciplined example does at least three scientific jobs at once. It tests whether the formal grammar can be applied without hand-waving. It reveals where an apparent theorem is only an interpretive gloss. And it gives editors and reviewers a compact way to see what would count as a counterexample before they invest in line-by-line reconstruction.

That is why the present section is included inside the master monograph rather than hidden in an outreach note. A large theory becomes more reliable when the examples are good enough to expose its failure modes, not when the examples are removed in the name of austerity.

**Bridge to the next chapter.** The next three chapters shift from interpretive checks to closure discipline: how promoted packets are operationalized, how empirical replay is bounded, and how proof machinery, competition, and compression stay synchronized with the science verdict.

| Observation or formal result                                                                                                                                                             | Immediate scientific consequence                                                                                                                                                                                                                                                                                                           | Release-level implication                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A nonempty admissible realization remains after the alleged collapse event.                                                                                                              | The death classification governed by Definition 12.1, Axiom 3.2, and Source Theorem 3 in Section 8 is not yet lawful.                                                                                                                                                                                                                      | Recheck threshold logic, collapse statement, and source witness used in the positive body.                                        |
| A later live continuum appears after lawful death and can still be proved numerically identical to the dead original continuum.                                                          | The irreversibility route fixed by Axiom 3.3, Definition 12.6, Source Corollary 3.2, and Lemma 3 in Section 8 is undermined.                                                                                                                                                                                                               | Revise or withdraw the irreversibility and rebirth-control branch; revise Theorem A only if the Lemma 2 route is also challenged. |
| Residue can be shown to preserve full live identity rather than only structural aftermath.<br>The journal core requires an excluded lift-heavy theorem family to sustain its main claim. | The classification boundary fixed by Definition 12.5, Lemma 2, and Theorem A in Section 8 is too weak or too broad as stated.<br>This schematic family-level objection targets the excluded lift-heavy theorem family marked in Figure 1; if that family is required, the extraction violates the support boundary stated in this section. | Rewrite the residue boundary before public release.<br>Narrow the article or restore support explicitly in the master first.      |
| $K_{11}$ and $K_{12}$ are shown reducible without loss to $K_{10}$ .                                                                                                                     | The upper-level doctrine stated here and in Section 7 is overstated under its own preservation test.                                                                                                                                                                                                                                       | Demote the hierarchy or re-prove the necessity of the upper layers.                                                               |

Table 3: Examples of observations or formal objections that would genuinely change the scientific status of the present release. The table is meant to preserve falsifiability, not to immunize the framework against criticism.

## A Core 1.3 Source-Audit Appendix

This appendix records the principles by which the 1.3 release remains source-first while still operating as a publication program rather than as a static archive.

### A.1 Why an appendix is still needed

The main body of the monograph is written to be read. The source-audit machinery is written to be checked. Those are related but non-identical tasks. If the full provenance burden were pushed into the main body, the book would become harder to read. If it were omitted entirely, the reader would need to trust a hidden workflow. The appendix therefore carries the discipline without replacing the main narrative.

### A.2 Witness classes

The release uses several witness classes:

- the public LaTeX corpus as the source-native textual witness,
- the archival compiled PDF as the stable page-and-line witness,
- the bounded journal-core extraction as the outward review-focused derivative witness,
- and companion packets as support witnesses for chronology, theorem fate, and release packaging.

None of these witnesses should be mistaken for the others. The source witness is not the same as the release manifest. The release manifest is not the same as the theorem. The journal-core article is not the same as the monograph. The 1.3 publication line remains healthy only when these roles stay distinct.

### A.3 Questions that the appendix answers

This appendix is meant to answer concrete review questions.

- From which public source stack is the present volume built?
- Which outward article is derived from it?
- Why is the monograph larger than the journal-core article?
- What justifies calling the volume 1.3 rather than simply republishing 1.2?
- How should later ESTRA and channel projections interpret the distinction between formal master, review article, and support packet?

The release is stronger when those questions can be answered inside the volume rather than by pointing the reader to an unrelated dashboard.

### A.4 Current boundary

Core 1.3 should therefore be read as a source-first enlargement and clarification of the formal publication shell. The monograph does not pretend that every future projection is already proved. The journal core does not pretend to be the whole theory. The companion packets do not pretend to be the manuscript. That three-way distinction is the governing boundary of the current release.

## B Journal-Core Extraction Bridge

This appendix explains how the journal-core article relates to the master monograph.

## **B.1 Why the journal-core exists**

Some channels require a bounded, faster-reading scientific artifact. Editors, reviewers, and external readers often need a paper-sized object whose defended claim can be read in one sitting. The journal-core exists for that purpose. Its job is not to replace the monograph but to provide a narrow article that remains anchored to the same scientific truth conditions.

## **B.2 What the journal-core may omit**

The journal-core may compress:

- extensive chapter-scale didactic explanation,
- large parts of the domain atlas,
- broad module coverage,
- detailed appendical inventories,
- and some of the monograph-level pedagogical scaffolding.

It may not silently change:

- theorem scope,
- explicit nonclaims,
- claimed source basis,
- figure meaning,
- or the relation between inherited corpus and present defended claim.

## **B.3 What the monograph guarantees to later artifacts**

Because the journal-core is extracted from the present volume, later owner-facing and public channels do not need to improvise their own understanding of the science. They can derive from the master monograph when they need depth, from the journal core when they need bounded exposition, and from the companion layer when they need support evidence.

This is the practical publication value of the monograph. It is not merely larger than the article. It is the document that prevents every later projection from becoming a separate interpretive exercise.

## C Reviewer Navigation Matrix and Audit Checklist

Start with the row that matches the question or concern, inspect the primary route, and use the verification column as the pass/fail test. This appendix uses three canonical artifact terms:

- *Master Monograph*: the English flagship volume. Later uses of *Monograph* are shorthand for this same artifact.
- *Journal Core*: the shorter bounded extraction from the monograph.
- *Release*: the outward package that also contains supporting surfaces, PDFs, assets, and companion dossiers.

Capitalisation marks the canonical glossary terms above; lowercase uses are generic prose shorthand unless explicitly styled as one of those terms. The following class labels are local definitions for this appendix rather than additional canonical glossary terms. Concrete artifact and witness classes are as follows:

- *Surfaces* are the release-bundled `editorial/*_latest.json` projections.
- *Companion dossiers* are the generated dossier packages under `editorial/dossier_packages/`.
- *Assets* are the released figures and tables under `assets/`.
- *Archival PDF witness*: the prior Core 1.2 PDF named in the journal-core provenance route in Appendix B: *Ontology of Continua — Core v1.2.0*.

### C.1 Review questions, compact route, and decisive checks

#### Core theorem route and scope.

| Reviewer question                                                       | Primary route to inspect | Verification cue                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the central defended result?                                    | Abstract; Sections 2, 8  | Theorem A: bounded post-collapse identity classification; no general lift theorem.                                                                                                                                                                                                                                                       |
| Where does the proof-bearing route begin?                               | Sections 2, 8            | Verify that Axiom 3.2 with Definitions 12.1, 12.3, and 12.4 supports Source Theorem 3; that Source Theorem 3 supports Lemma 1; that Lemma 1 with Definition 12.5 supports Lemma 2; and that Lemma 2 supports Theorem A. Inspect Axiom 3.3, Definition 12.6, and Source Corollary 3.2 separately as the companion rebirth-control branch. |
| Is collapse defined formally or rhetorically?                           | Sections 2, ??           | Admissible-state emptiness, $k \rightarrow 0$ , and threshold failure.                                                                                                                                                                                                                                                                   |
| What exactly is residue?                                                | Sections ??, 9           | Post-collapse structural trace; no continuing live identity.                                                                                                                                                                                                                                                                             |
| What exactly is rebirth?                                                | Sections ??, 9           | Later live continuum grounded in residue; numerically new.                                                                                                                                                                                                                                                                               |
| Why is irreversibility necessary?                                       | Sections 2, 8            | Blocks same-entity restoration after lawful death.                                                                                                                                                                                                                                                                                       |
| Could the main theorem depend on an excluded lift-heavy theorem family? | Section 8; Appendix B    | No excluded lift-heavy theorem family is used in the bounded proof route.                                                                                                                                                                                                                                                                |
| Why is the monograph larger than the journal core?                      | Section 6; Appendix B    | Didactic shell, formal architecture, and repair logic live here.                                                                                                                                                                                                                                                                         |
| Why is the journal core smaller without becoming weaker?                | Appendix B               | One-way extraction; theorem scope unchanged.                                                                                                                                                                                                                                                                                             |
| Why is the hierarchy fixed as $K_0, \dots, K_{12}$ ?                    | Sections 7, ??, 9        | Upper levels carry substantive higher-order roles.                                                                                                                                                                                                                                                                                       |

### Audit, provenance, and review posture.

| Reviewer question                                | Primary route to inspect   | Verification cue                                                                                                                                                                 |
|--------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What would falsify the upper-level doctrine?     | Sections 9, ??             | Lossless reduction of $K_{11}$ and $K_{12}$ to $K_{10}$ .                                                                                                                        |
| How do proof-oriented statement classes differ?  | Sections 7, 8              | Axiomatic, derived, structural, interpretive, empirical, and frontier rows keep separate burdens; editorial-bridge language remains packaging support rather than a proof class. |
| What is merely interpretive?                     | Sections ??, 6, 9          | Pass only if interpretive language stays outside the proof route and is not promoted into theorem, replay, or practical-use authority.                                           |
| What is the exact source base?                   | Section 6; Appendix A      | Source corpus, SPOT surface, and archival PDF witness are separate objects.                                                                                                      |
| Why is ORCID foregrounded?                       | Title page; Section 6      | Stable scholarly identity.                                                                                                                                                       |
| What is the role of companion dossiers?          | Section 6; Appendix B      | Support and provenance exposure, not replacement authority.                                                                                                                      |
| How should a non-specialist enter the monograph? | Sections 1, 9              | Didactic ladder before proof audit.                                                                                                                                              |
| Where are the most likely points of critique?    | Section 8; Appendix B      | Collapse semantics, irreversibility, upper levels, extraction discipline.                                                                                                        |
| What would count as a decisive release failure?  | Sections 9, ??, 6          | Continued identity after admissible-state loss, or reliance on an excluded theorem family.                                                                                       |
| What remains outside scope?                      | Sections ??, 6; Appendix B | Pass only if excluded theorem families, frontier rows, and projection targets remain outside the positive proof body.                                                            |

## C.2 Compact reading order by reviewer role

**Rapid critical reviewer.** Read the abstract, theorem roadmap, relevant theorems and corollaries, the worked examples, and the journal-core bridge appendix. Return to this matrix only to choose a new route after that first pass. This route should reveal immediately whether the release is too large, too narrow, or too broad in scope.

**Formal mathematical reviewer.** Read the model, results, axioms, operators, collapse/rebirth material, hierarchy chapter, notation appendix, and source-audit chapter. The decisive question is whether the derived bounded claim can be reconstructed from the stated source-native rows without informal supplementation.

**Generalist scientific reviewer.** Read the introduction, reader guide, theorem roadmap, worked examples, discussion, conclusion, and then return to the formal chapters selectively. The decisive question here is whether the volume explains itself honestly enough that scientific judgment can precede local notation mastery.



## D OC 1.3.1 Science Delta Appendix

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## E OC 1.3.1 Critique Closure Appendix

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audience
- critique\_id=CRITIQUE::PUBLIC\_REPO\_SIMULATION\_ABSENCE; processing\_class=REAL\_PACKAGING\_WEAK  
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